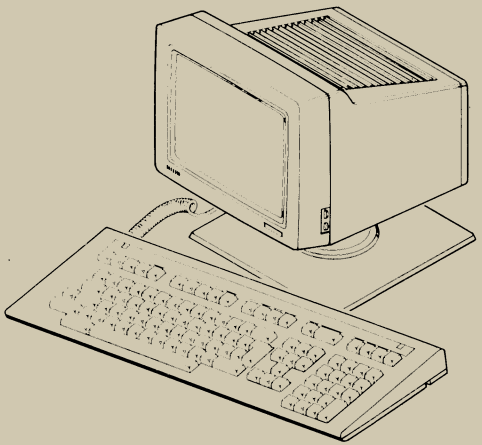
EK-VT340-PS-002

VT340

Pocket Service Guide



digital™

EK-VT340-PS-002

VT340

Pocket Service Guide

Prepared by Educational Services
of
Digital Equipment Corporation

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DECtalk	ReGIS	VT241
DECUS	RSTS	VT330, VT340
DECwriter	RSX	Work Processor
DIBOL	SSU	

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VT340 VIDEO TERMINAL

The VT340 is a text and color graphics video terminal that is compatible with the VT241. The VT340 supports different versions of Digital's LK201 corporate keyboard, for different languages.

The VT340 uses a four-plane bitmap to display 16 colors at a time, from a palette of 4096 possible colors. The monitor has a 332 mm (13 inch) antiglare screen. The VT340 can display 24 rows of text in 80 or 132 columns. The twenty-fifth display line is a status line. The terminal's built-in tilt-swivel base lets the user adjust the screen for viewing comfort.

The VT340 has five interface connectors.

Connector	Connects the VT340 to...
Comm1 RS232 (25-pin)	A primary host computer, directly or indirectly (through a terminal server or modem).
Comm1 DEC-423 (6-pin)	A primary host computer, directly or indirectly (through a terminal server).
Comm2 DEC-423 (6-pin)	A secondary host computer, directly or indirectly (through a terminal server).
Printer DEC-423 (6-pin)	A printer.
Mouse/tablet (7-pin micro-DIN)	A mouse or graphics tablet.

The VT340 operates on full-duplex, asynchronous communication lines.

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ABOUT THIS GUIDE

This guide describes how to service the VT340 color video terminal in the field. The guide covers the following topics.

Testing and troubleshooting
the field replaceable units (FRUs)
Removing and replacing FRUs
Aligning the video monitor
Using set-up

Appendices provide information on the following topics.

Related documentation
Troubleshooting the Session Support Utility
FRU exploded view diagrams
Cables
Physical/functional diagram

WARNINGS, CAUTIONS, AND NOTES

The warnings, cautions, and notes in this guide have specific purposes.

WARNING	Contains information to prevent personal injury.
CAUTION	Contains information to prevent damage to equipment.
NOTE	Contains general information you should be aware of.

VT340 VIDEO TERMINAL

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Printer DEC-423 (6-pin)	A printer.
Mouse/tablet (7-pin micro-DIN)	A mouse or graphics tablet.

The VT340 operates on full-duplex, asynchronous communication lines.

TOOLS AND EQUIPMENT

You need the following tools and equipment to service the VT340 video terminal.

Tools and Equipment	Part Number
Phillips screwdriver, number 2	29-13510-00
Anode discharge tool	29-24717-00
Video alignment tool	29-24746-00
Tuning wand	29-23190-00
Metric measuring tape	29-25342-00
Loopback connectors	
Host EIA (25-pin)	12-15336-00
Modular jack (6-pin)	12-25083-01
Micro-DIN (7-pin)	12-25628-01
Color video service kit (without power pack)	A2-S0099-00
Power packs	
90 to 110 Vac, 50 Hz	29-25448
104 to 126 Vac, 60 Hz	29-25449
198 to 242 Vac, 50 Hz	29-25450

RECOMMENDED SPARES LIST

The following is the recommended spares list for the VT340.

Spares	Part Number
Terminal control module	54-16965-01
Video amp module	54-16963-01
Control/bracket assembly	70-22618-02
Inlet/switch assembly	70-23685-01
PS/monitor and arc protection assembly	70-24516-01
Rear enclosure/base assembly	70-23687-01
ROM cartridge assembly	54-16967-04
CRT/bezel/chassis assembly	
(Northern Hemisphere)	70-23688-01
(Southern Hemisphere)	70-23688-02
Rear panel	74-33422-01

Spares	Part Number
Keyboards	
North American/U.K.	LK201-NA
North American/U.K. (WPS)	LK201-PA
Belgian	LK201-LB
Danish	LK201-RD
Dutch	LK201-LH
Finnish	LK201-NF
French	LK201-LP
French Canadian	LK201-LC
German/Austrian	LK201-LG
Italian	LK201-LI
Norwegian	LK201-RN
Portuguese	LK201-LV
Spanish	LK201-LS
Swedish	LK201-LM
Swiss (French)	LK201-LK
Swiss (German)	LK201-LL
Options	
Mouse	VSXXX-AA
Graphics tablet	VSXXX-AB

1 TESTING AND TROUBLESHOOTING

1.1 GENERAL

This chapter describes how to run the self-tests built into the VT340 video terminal. The chapter also lists error codes for the tests and describes how to troubleshoot general problems.

1.2 USING SELF-TESTS

The VT340 has seven self-tests to help you isolate failures to the terminal's field replaceable units (FRUs). You run these tests from the terminal's Diagnostic Set-Up screen (Paragraph 1.4). The power-up self-test also runs each time you turn the terminal on.

If a test finds a faulty FRU, adjust or replace the faulty unit. Then repeat the tests in this chapter, to ensure the terminal operates correctly. Appendix C shows each FRU.

1.3 POWER-UP SELF-TEST

This test runs automatically each time you turn the terminal on. The test checks the terminal's internal memory, keyboard, and video circuitry. The test also makes sure the communication ports, printer port, and locator device port are operating correctly. The loopback self-tests on the Diagnostic Set-Up screen perform a more complete test of each port.

A successful power-up self-test ends as follows (Figure 1-1).

1. The keyboard's LED indicators are off.
2. The keyboard makes the bell tone sound.
3. A "VT340 OK" message appears in a rectangle on the screen, With the firmware revision level and a set-up screen copyright message. This message disappears when

   - the terminal receives any character except XON, XOFF, or NULL.
   - you press any key.
   - you leave the terminal on but inactive for 30 minutes.

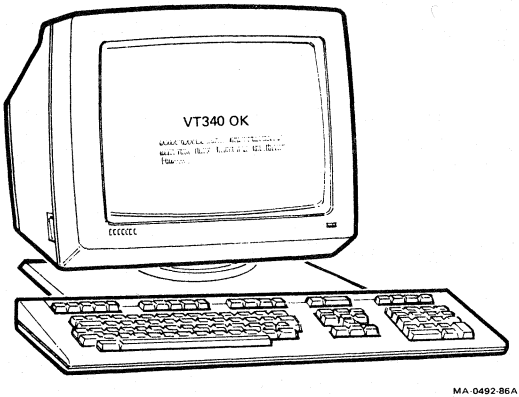


Figure 1-1 Successful Power-Up Self-Test Display

If the Test Finds an Error

First look at the screen. The terminal will display one or more error messages. Each error message starts on a new line at the left margin of the display. Paragraph 1.7 describes the error messages.

If the screen display is faulty, check the keyboard LED indicators. They also display the error code.

If other problems occur, see Paragraph 1.8. It lists some common operating problems, their probable causes, and suggested solutions.

1.4 DIAGNOSTIC SET-UP

You use the Diagnostic Set-Up screen (Figure 1-2) to run one or more of the terminal's seven self-tests. Paragraph 1.5 describes each test.

You can run the five loopback tests and the power-up test in any combination. Each test runs until it is complete, or until the terminal detects an error. You must run the screen tests separately from the loopback tests and power-up test.

NOTE
If you are unfamiliar with the terminal's set-up screens, see Chapter 4.

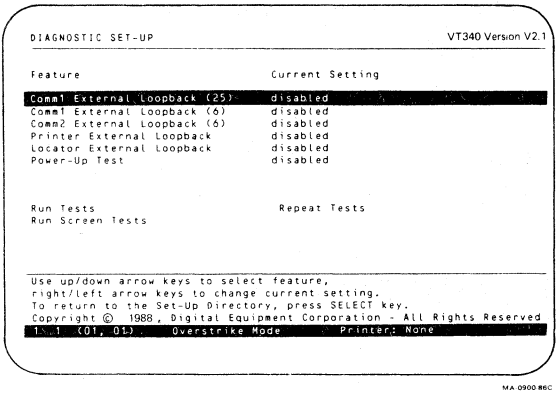


Figure 1-2 Diagnostic Set-Up Screen

The top half of the Diagnostic Set-Up screen has two columns.

```
Feature        Current Setting
```

Feature Column
This column lists the individual loopback tests and the power-up test.

Current Setting Column
This column shows you if a test is enabled or disabled. You must enable a test before you can run it. Initially, all tests are disabled.

How to Change a Test's Current Setting
Use the ↑ and ↓ keys to move the cursor to the selected test in the feature column. Use the ← and → keys or **Enter** key to change the test's current setting (enabled or disabled).

How to Run a Self-Test
You can run one or more self-tests as follows. You can run the tests one time or repeatedly. Before you run a loopback test, you must connect a loopback connector to the selected port (Paragraph 1.5.1). To run the screen tests, see Chapter 3.

1. Press **Set-Up**. The Set-Up Directory appears.
2. Use the arrow keys to move the cursor to **Diagnostic Set-Up**. Press **Enter** to display the Diagnostic Set-Up screen.
3. Use the arrow keys to enable each test you want to run.
4. Use the arrow keys to move the cursor to **Run Tests** or **Repeat Tests**.

   Run Tests performs each enabled test one time.

   Repeat Tests performs each enabled test continuously.
5. Press **Enter** to start the self-tests.

   NOTE
   The continuously running test ends when an error occurs or when you turn off the power. The keyboard does not make a bell tone sound during a continuously running test.

1.5 DIAGNOSTIC SELF-TESTS

You can select seven different self-tests from the Diagnostic Set-Up screen. The first five tests are loopback tests that check the terminal's three communication ports, the printer port, and the locator device port.

The sixth test is the power-up self-test. The last test is a set of screen patterns that field service personnel use to adjust the video display (Chapter 3).

1.5.1 Loopback Tests

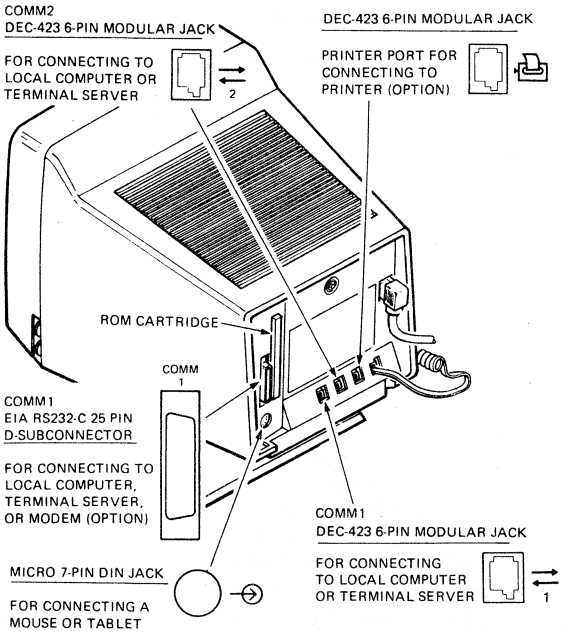
All five loopback tests work the same way. You plug a loopback connector into the port you want to test. The loopback connector connects the port's transmit and receive data lines.

When you run the test, the terminal sends a predefined set of characters on its transmit line and receives them on its receive line. Then the terminal compares the output characters to the input characters. If the characters do not match, an error message appears on the screen.

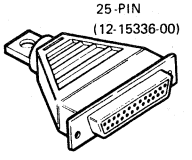
You must use a loopback connector, or the test will fail. Use the following loopback connectors.

Loopback Connector	Part Number
Host EIA port (25-pin)	12-15336-00
Modular jack (6-pin)	12-25083-01
Micro-DIN (7-pin)	12-25628-01

Figure 1-3 shows the five VT340 ports and the three loopback connectors. Note that the VT340 has two Comm1 connectors.



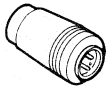
LOOPBACK CONNECTORS



FROM PIN	TO PIN	TO PIN
2	3	15
4	5	8
6	20	22
12	19	17



FROM PIN	TO PIN
1	6
2	5
3	4



FROM PIN	TO PIN
2	3

MA-0622-86

Figure 1-3 VT340 Ports and Loopback Connectors

How to Run a Loopback Test

1. Turn the terminal off (0).
2. Disconnect the cable from the port you are testing.
3. Connect the appropriate loopback connector to the port.

Test	Loopback Connector
Comm1 (25-pin)	12-15336-00
Comm1 (6-pin)	12-25083-01
Comm2 (6-pin)	12-25083-01
Printer (6-pin)	12-25083-01
Locator (7-pin)	12-25628-01

4. Turn the terminal on (1).
5. Use the Diagnostic Set-Up screen to run the test.

A successful test ends as follows.

1. The keyboard's LED indicators are off.
2. The keyboard makes a bell tone sound.
3. A "VT340 OK" message appears on the screen.

If the Test Finds an Error
Your screen may display one of the following messages. See Paragraph 1.7 for details.

Comm1 25-pin test

```
VT340 Comm1 Port Data Error -- 2

VT340 Comm1 Port Control Error -- 3
```

Comm1 6-pin test

```
VT340 Comm1 DEC-423 Data Error -- 5
```

Comm2 6-pin test

```
VT340 Comm2 DEC-423 Data Error -- 6
```

Printer 6-pin test

```
VT340 Printer Port Error -- 7
```

Locator 7-pin test

```
VT340 Locator Port Error -- 8
```

1.5.2 Power-Up Test

This test is the same as the power-up self-test described in Paragraph 1.3. If you select this test, the continuously running test ends when an error occurs or when you turn off the power.

1.5.3 Screen Tests

These tests are a series of calibration and alignment displays. You use these patterns to adjust the display height, width, and linearity. You do not need a loopback connector for these tests.

See Chapter 3 for the procedures used with these tests.

1.6 PRINTER PROBLEMS

If a printer connected to a VT340 does not print, use the following steps to isolate the problem to the printer or terminal.

1. Run the power-up self-test (Paragraph 1.3). If the terminal passes the test, go to step 2.
2. Run the printer external loopback self-test (Paragraph 1.5.1). If the terminal passes this test, go to step 3.

   NOTE
   If your terminal passes the above tests, it is running correctly. The problem is probably not in the terminal.

3. Test the printer. See your printer's pocket service guide for the correct procedures. If your printer is operating correctly, go to step 4.

4. Check the following set-up features to make sure your terminal and your printer are compatible. If you are unfamiliar with the terminal's set-up screens, see Chapter 4.

- Check the **Printer Assignment** feature in the Global Set-Up screen. Make sure the printer is assigned to the session you are in. If the printer is assigned to the other session, the printer will not print from your session.
- Check the printer's status on the status line in set-up.

Printer Status	Meaning
Ready	The printer is available to print.
Not Ready	The printer port is not available to print.
None	The printer is turned off or not connected to the printer port.
Auto Print	When you or the host end a line with a carriage return, the terminal sends the line to the printer.
Controller	Information from the host goes directly to the printer.
Busy	The printer port is performing another operation (such as **Local Print**).
Not Assigned	The printer port is not assigned to the active session.

- Check the following features in the Printer Set-Up screen. The settings for the printer and terminal must match.

  Print speed (75 to 19.2K baud)
  Character format (8- or 7-bit; even, odd, or no parity)
  Stop bits (1 or 2)
  Flow control (XON/XOFF, DSR/DTR)

5. Make sure you are using the correct cable between your terminal and printer (Appendix D). Make sure the connections are secure at both ends of the cable.
6. Put your terminal in the local state by using the **On Line/Local** feature in the Global Set-Up screen. Then leave set-up and type some characters on the screen.
7. Press the **Local Print** key. The data on your screen should print out if the printer is operating correctly. If the data does not print out, go back to step 3.

1.7 SELF-TEST ERROR CODES

If your screen displays an error message, see Table 1-1. It lists the field replaceable unit (FRU) you must replace to correct the problem.

If your VT340 fails the power-up self-test, see the troubleshooting chart, Table 1-2.

If there is a problem with the screen display, check the keyboard's LED indicators. The LED indicators always display the error code for the test currently running. This test must run successfully before your terminal clears the LED error code.

If the LEDs flash or stay on during the test, replace the terminal control module (Paragraph 2.5).

See Table 1-2 for other troubleshooting procedures.

1.7.1 NVR Error — 1 Code

If a "VT340 NVR Error — 1" message appears when you turn on VT340, there is a problem with the terminal's nonvolatile memory (NVR) that stores the factory-default set-up settings. You can save the factory-default settings in NVR as follows. This procedure applies to single or dual sessions.

1. Press Set-Up. The Set-Up Directory appears.
2. Move the cursor to **Recall Factory Default Settings** and press **Enter**.
3. Move the cursor to **Save Current Settings** and press **Enter**.
4. Move the cursor to **User-Defined Key Set-Up** and press **Enter**. The User-Defined Key Set-Up screen appears.
5. Move the cursor to **Save User-Defined Keys** and press **Enter**.
6. Press **Select** to return to the Set-Up Directory.
7. Press **Next Screen** to display the Global Set-Up screen.
8. Move the cursor to **Dual Terminal** and press **Enter** to enable the dual sessions feature.
9. Press **Set-Up** to leave set-up.
10. Press **Switch Session** (F4) to move to the second session.
11. Press **Set-Up** to display the Set-Up Directory again.
12. Move the cursor to **Recall Factory Default Settings** and press **Enter**.
13. Move the cursor to **Save Current Settings** and press **Enter**.
14. Press **Set-Up** to leave set-up.
15. Turn the terminal off, then on. If the "VT340 NVR Error — 1" message still appears on the screen, the terminal has a hardware failure. See Table 1-1.

Table 1-1 VT340 Display Error Codes

Error Message*	Solution†
NVR Error — 1	1. Recover from the NVR (1.7.1). 2. Replace the terminal control module (2.5).
Comm1 Port Data Error — 2	Replace the terminal control module (2.5).
Comm1 Port Control Error — 3	Replace the terminal control module (2.5).
Keyboard Error — 4	1. See if the keyboard is plugged in. 2. Turn the VT340 off, then on. 3. If the error continues, replace the keyboard (2.14). 4. If the error still continues, replace the PS/monitor and arc protection assembly (2.7).
Comm1 DEC-423 Data Error — 5	Replace the PS/monitor and arc protection assembly (2.7).
Comm2 DEC-423 Data Error — 6	Replace the PS/monitor and arc protection assembly (2.7).
Printer Port Error — 7	Replace the PS/monitor and arc protection assembly (2.9).
Locator Port Error — 8	1. See if the mouse/tablet is plugged in. 2. Turn the VT340 off, then on. 3. If the error continues, replace the mouse/tablet (2.15). 4. If the error still continues, replace the terminal control module (2.5).

* All error messages start with "VT340".

† Paragraph numbers are in bold type.

1.8 TROUBLESHOOTING GENERAL PROBLEMS

Table 1-2 lists some common operating problems, their probable causes, and suggested solutions.

Table 1-2 VT340 Troubleshooting Chart

Symptom	Probable Cause	Suggested Solution*
No VT340 OK display, power indicator off, no bell tone.	VT340 not plugged in, or no power at wall outlet.	Plug in the VT340, or try another wall outlet.
	AC power cord	Check for an open line or shorts.
	Voltage select jumper on wrong pin.	Check that the voltage select jumper is configured correctly on the PS/monitor and arc protection assembly.
	Inlet/switch assembly	Check the connections on the PS/monitor and arc protection assembly.
	PS/monitor and arc protection assembly	Replace the PS/monitor and arc protection assembly (2.7).
Power indicator flashes.	ROM cartridge	Check that the ROM cartridge is installed.
	Terminal control module	Replace the terminal control module (2.5).
Keyboard LEDs flash or stay on.	Terminal control module	Replace the terminal control module (2.5).
Compressed raster (white lines) appears, but no video display.	Video cable connections	Check the connections from the PS/monitor and arc protection assembly to the CRT yoke.
	Video amp module	Replace the video amp module (2.8).
Raster appears, but no video display.	Video amp module	Check connectors and reconnect.
		Replace the video amp module (2.8).

* Paragraph numbers are in bold type.

Table 1-2 VT340 Troubleshooting Chart (Cont)

Symptom	Probable Cause	Suggested Solution*
CRT filaments not lit.	CRT/bezel/chassis assembly	Replace the CRT/bezel/chassis assembly (2.13).
VT340 OK display present, power indicator off.	LED assembly	Check the connection to the PS/monitor and arc protection assembly (2.7).
		Replace the LED assembly (2.11).
Screen display distorted.	Monitor is out of alignment.	Perform the alignment procedures (Chapter 3).
	Monitor circuit	Replace the PS/monitor and arc protection assembly (2.7).
	Terminal control module	Replace the terminal control module (2.5).
	CRT/bezel/chassis	Replace the CRT/bezel/chassis (2.12).
No bell tone.	Warning bell is disabled.	Set **Warning Bell** to "high" or "low" in Keyboard Set-Up.
	Keyboard speaker faulty.	Replace the keyboard (2.14).
Different characters appear on screen than were typed in local.	Alternate character set selected.	Use **Recall Saved Settings** in the Set-Up Directory.
	Keyboard	Replace the keyboard (2.14).
	Terminal control module	Replace the terminal control module (2.5).
Different characters appear on screen than were typed while on-line with host. (Terminal works in local).	Transmit and receive speeds are set wrong.	Set speeds to match host (Communications Set-Up).
	Bits per character or parity setting is wrong.	Set **Character Format** to match host (Communications Set-Up).
	Stop Bits setting is wrong.	Set **Stop Bits** to match host (Communications Set-Up).

* Paragraph numbers are in bold type.

Table 1-2 VT340 Troubleshooting Chart (Cont)

Symptom	Probable Cause	Suggested Solution*
Screen display does not scroll. **Hold Session** indicator is on.	Screen display is locked.	Press the **Hold Session** key to release.
Terminal appears locked, does not respond to data from host.	—	Clear the terminal with the **Clear Communications** field in the Set-Up Directory.
Screen goes blank after successful power-up, then is inactive for one-half hour. Power indicator is on.	**CRT Saver** feature in Global Set-Up is enabled.	Press any key to reactivate screen.
Messages are incomplete.	XON/XOFF not selected.	Set **Transmit Flow Control** to "XON/XOFF" (Communications Set-Up).
	Comm port connections	Check cables at comm ports.
	CPU module	Replace the CPU module (2.5).
Terminal does not respond to an escape sequence.	Incorrect **Terminal Mode** selected.	Check the setting of the **Terminal Mode** feature in General Set-Up.

* Paragraph numbers are in bold type.

2 REMOVING AND REPLACING FRUs

2.1 GENERAL

This chapter shows you how to remove and replace VT340 field replaceable units (FRUs). Appendix C contains exploded view drawings that provide the location, part name, and part number of each FRU.

> **CAUTION**
> **Use a static protection kit (PN 29-11762-00) when handling any internal components.**

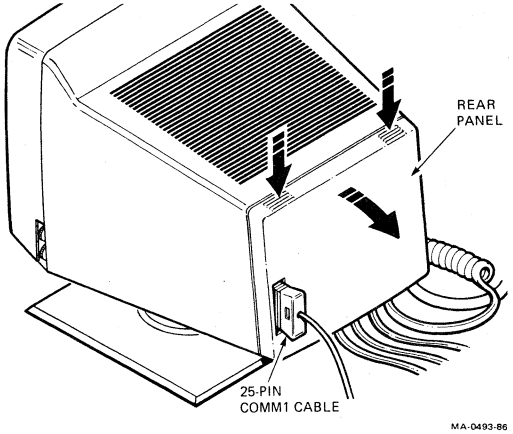
2.2 REAR PANEL

Remove the rear panel as follows.

1. Grasp the top of the rear panel and pull toward you.
2. If there is a cable connected to the 25-pin Comm1 port, slide the panel down the cable and away from the terminal.

   Otherwise, lower the panel and remove from the terminal. Set the panel aside.

   To install the rear panel, reverse steps 1 and 2.



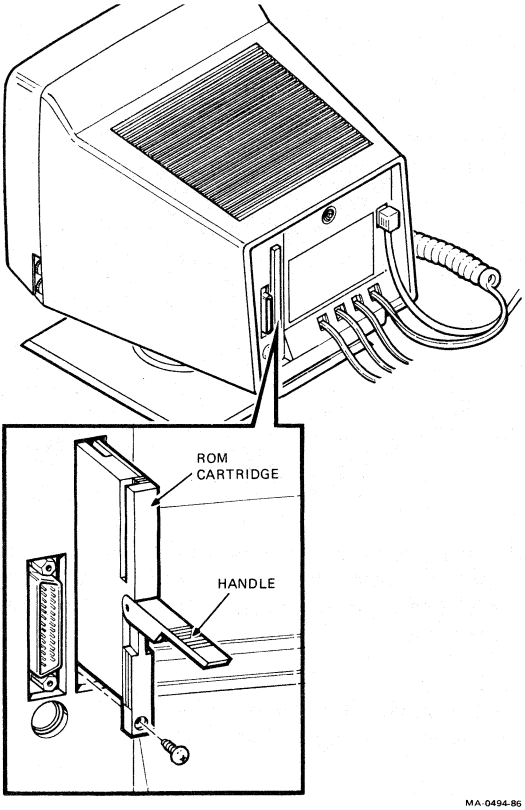
2.3 ROM CARTRIDGE

Remove the ROM cartridge as follows. First remove the following FRU.

Rear panel (Paragraph 2.2)

1. Turn the terminal's power switch off (0).
2. Lift the handle on the ROM cartridge.
3. Remove the phillips screw on the bottom of the ROM cartridge. This screw holds the ROM cartridge to the terminal control module.
4. To remove the ROM cartridge, grasp the handle and pull firmly.

To install the ROM cartridge, reverse steps 1 through 4.



MA-0494-86

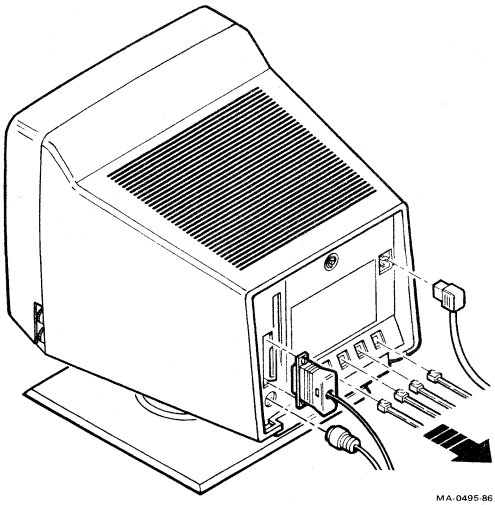
2.4 REAR ENCLOSURE/BASE ASSEMBLY

Remove the rear enclosure/base assembly as follows. First remove the following FRU.

Rear panel (Paragraph 2.2)

1. Unplug the power cord from the wall outlet first, then from the terminal.
2. Disconnect the keyboard cable from the terminal.
3. Disconnect all other cables from the rear of the terminal, including the following.

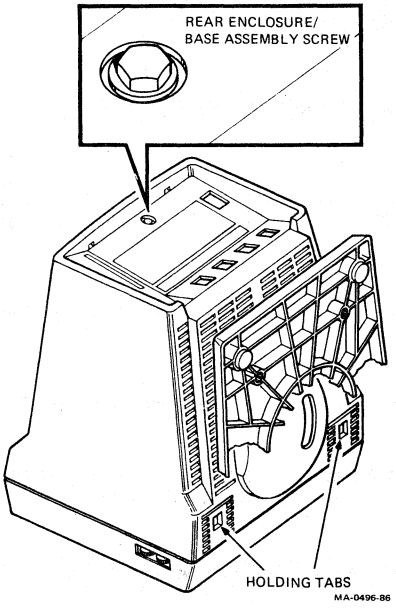
   - host port cable(s)
   - printer cable
   - locator device cable



4. Place a piece of paper on a flat working surface. Carefully place the monitor facedown on the paper. The paper prevents scratches on the monitor bezel.

   NOTE
   You must clean the face of the CRT after you service the terminal.

5. Use a 5/16 inch nutdriver to loosen the captive screw on the rear panel. This screw holds the rear enclosure/base assembly to the chassis.



WARNING
The next step exposes you to the CRT anode, which stores a high voltage. Use caution while the rear enclosure is off the terminal.

6. Remove the rear enclosure/base assembly by pressing the two holding tabs in and sliding the cover straight up.

To install the rear enclosure/base assembly, reverse steps 1 through 6.

CAUTION
When you install the rear enclosure/base assembly, carefully align the screw hole and holding tabs.

2.5 TERMINAL CONTROL MODULE

Remove the terminal control module as follows. First remove the following FRUs.

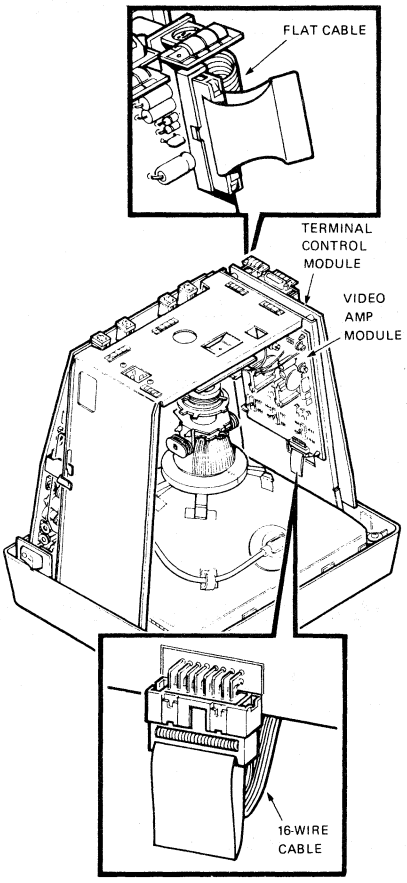
Rear panel (Paragraph 2.2).

ROM cartridge (Paragraph 2.3).

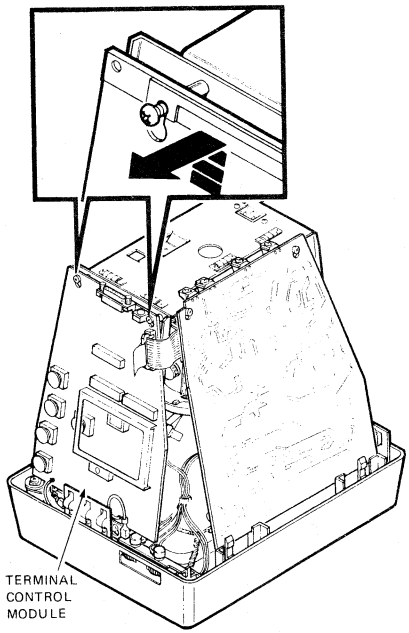
Rear enclosure/base assembly (Paragraph 2.4).

CAUTION
Use a static protection kit (PN 29-11762-00) when handling any internal components.

1. Disconnect the following cables on the terminal control module.
   - flat cable to the PS/monitor and arc protection assembly
   - 16-wire cable to the video amp module (This cable is on the rear of the terminal control module.)
2. Loosen the two phillips screws that hold the terminal control module to the chassis.


FLAT CABLE
TERMINAL CONTROL MODULE
VIDEO AMP MODULE
16-WIRE CABLE
MA-0206-88

3. Slide the module up and gently pull the module toward you. Feed the cable through the opening in the chassis.



To install the terminal control module, reverse steps 1 through 3. The terminal control module has two notches on the bottom. Place the notches on the legs of the chassis.

2.6 DISCHARGING THE CRT

Discharge the CRT as follows. First remove the following FRUs.

> **WARNING**
> **The following steps expose you to the CRT anode, which may store a high voltage.**

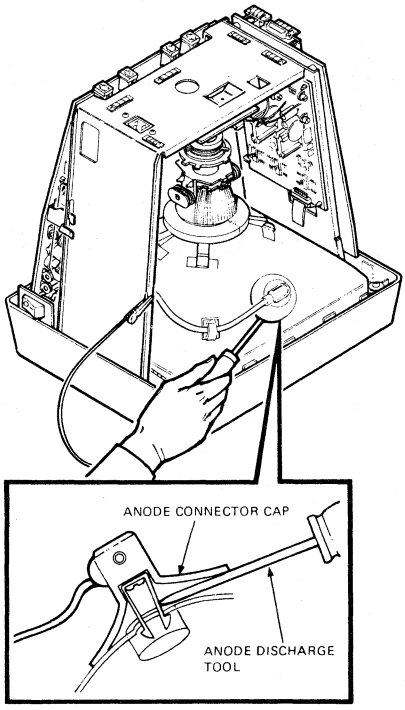
Rear panel (Paragraph 2.2).

Rear enclosure/base assembly (Paragraph 2.4).

1. Attach the clip end of the anode discharge tool (PN 29-24717-00) to the metal chassis.
2. Push the probe end under the soft plastic anode connector cap, until you feel the probe touching the anode connector.

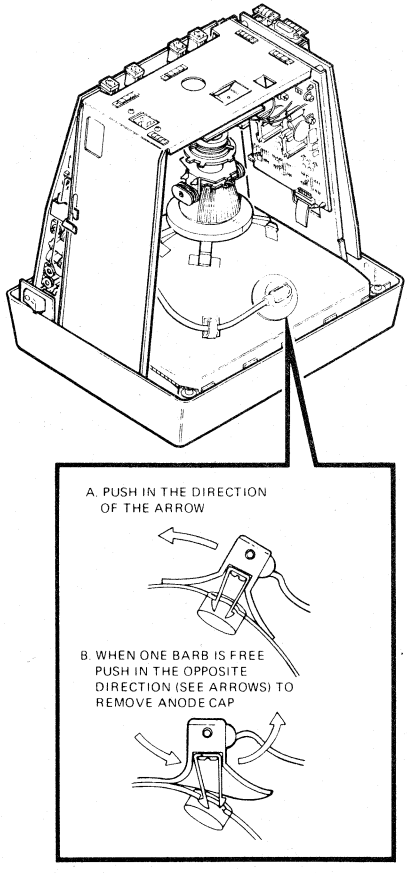
   CAUTION
   Do not scratch the glass of the CRT when discharging the anode.

3. Hold the probe against the anode connector for at least 3 seconds, then remove the probe.



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4. Remove the CRT anode wire from the cable clamp on the CRT.
5. Remove the CRT anode connector from the CRT, as shown.



2.7 POWER SUPPLY/MONITOR AND ARC PROTECTION ASSEMBLY

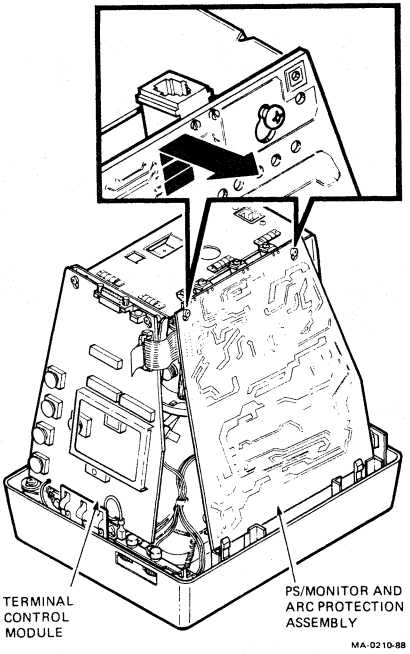
Remove the power supply (PS)/monitor and arc protection assemly as follows. First remove the following FRUs.

Rear panel (Paragraph 2.2).

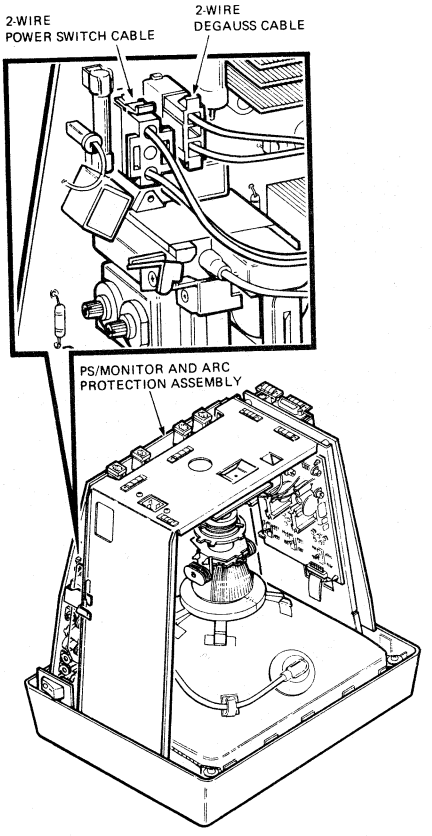
Rear enclosure/base assembly (Paragraph 2.4).

WARNING
You must discharge the CRT (Paragraph 2.6) before removing the PS/monitor and arc protection assembly.

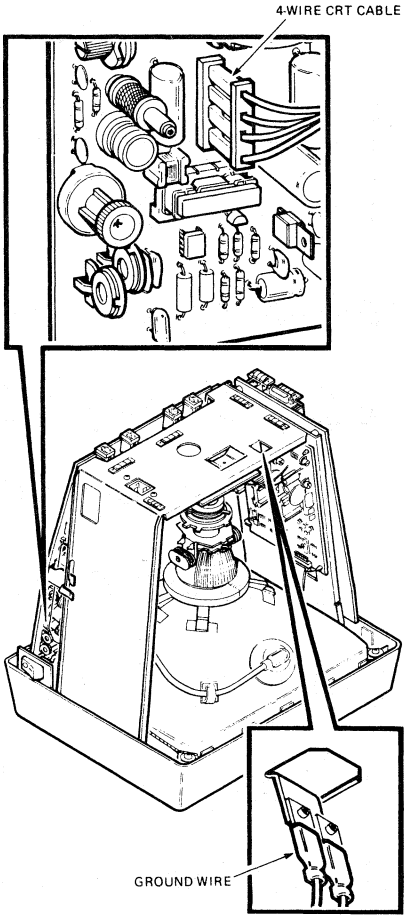
1. Disconnect the flat cable on the terminal control module. This cable goes to the PS/monitor and arc protection assembly.
2. Loosen the two phillips screws that hold the PS/monitor and arc protection assembly to the chassis.
3. Gently slide the module up and off the screws.



4. Disconnect the 2-wire internal degauss cable on the PS/monitor and arc protection assembly.
5. Remove the internal degauss cable and CRT socket cable from the cable clamp on the side of the chassis
6. Disconnect the 2-wire power switch cable on the PS/monitor and arc protection assembly.

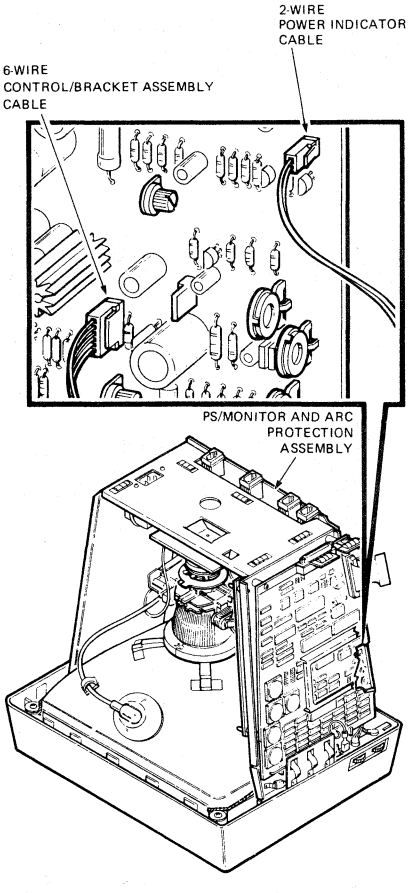


7. Disconnect the 4-wire CRT cable on the PS/monitor and arc protection assembly. This cable goes to the CRT yoke.
8. Disconnect the ground wire on the chassis that goes to the PS/monitor and arc protection assembly.



9. Disconnect the following cables on the PS/monitor and arc protection assembly. To disconnect a cable, press the tab under the connector.

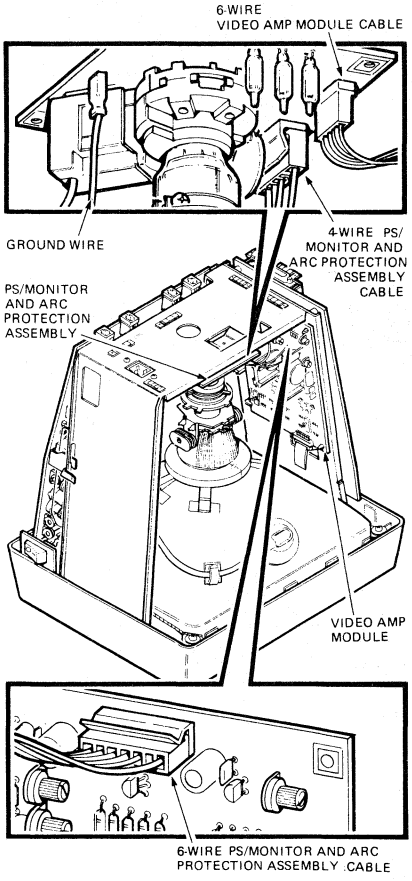
   - 2-wire cable to the power indicator
   - 6-wire cable to the control/bracket assembly



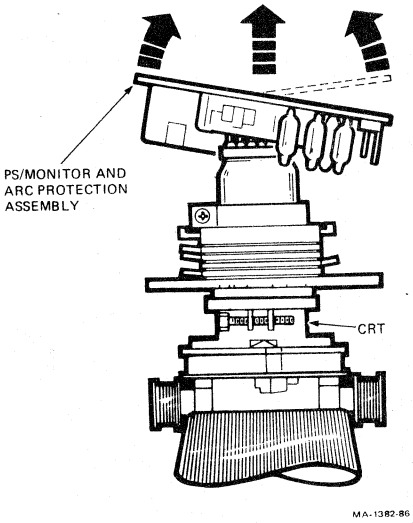
10. Part of the PS/monitor and arc protection assembly sits on the neck of the CRT. Disconnect the following wires from that part.

    - 4-wire cable
    - white ground wire
    - 3-wire cable to the video amp module

11. Disconnect the 6-wire cable on the video amp module. This cable comes from the PS/monitor and arc protection assembly.

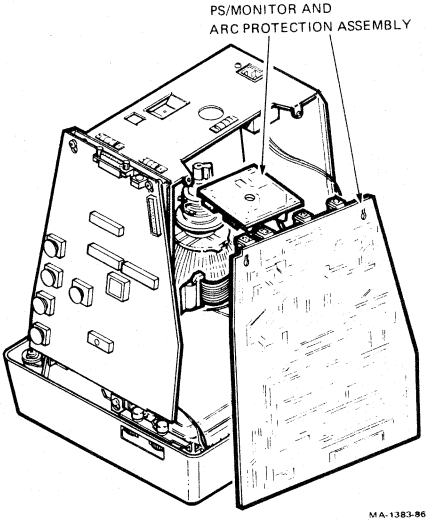


12. Carefully lift the PS/monitor and arc protection assembly off the neck of the CRT by using a gentle side-to-side motion.



MA-1382-86

13. Remove the PS/monitor and arc protection assembly from the chassis.



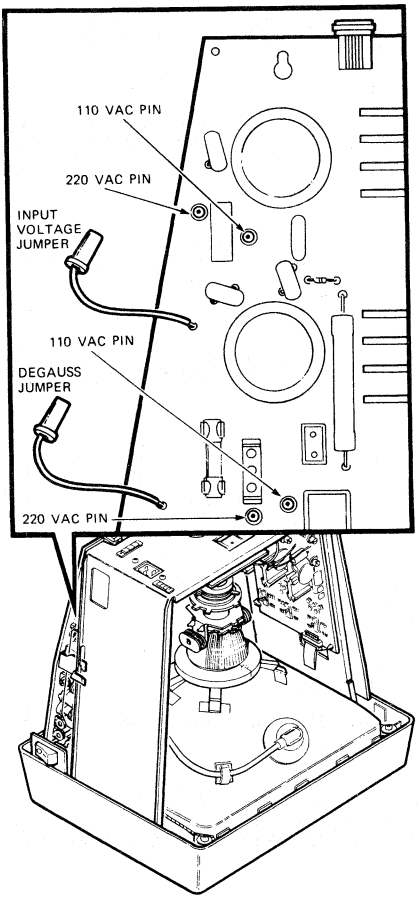
MA-1383-86

To install the PS/monitor and arc protection assembly, reverse steps 1 through 13.

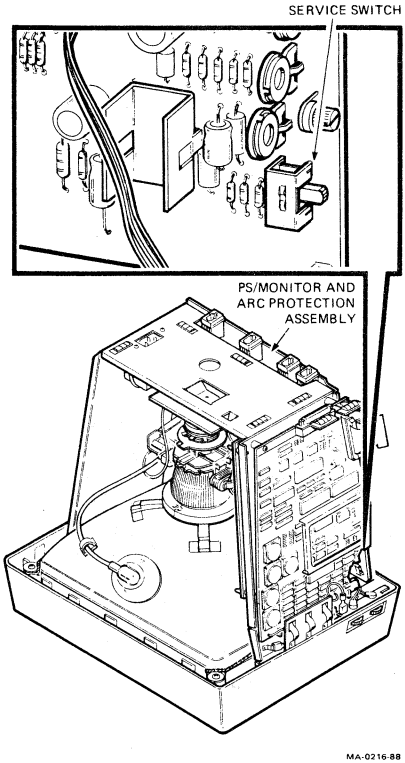
2.7.1 Input Voltage Selection

Make sure you set the jumpers on the PS/monitor and arc protection assembly to match the input voltage used.

1. For 110 V, place the input voltage select jumper on the 110 pin. For 220 V, place the jumper on the 220 pin.
2. For 110 V, place the degauss jumper on the 110 pin. For 220 V, place the jumper on the 220 pin.



NOTE
Make sure the service switch is between the two white lines on the PS/monitor and arc protection assembly. This setting indicates the terminal is in the operating state.



2.8 VIDEO AMP MODULE

Remove the video amp module as follows. First remove the following FRUs.

Rear panel (Paragraph 2.2).

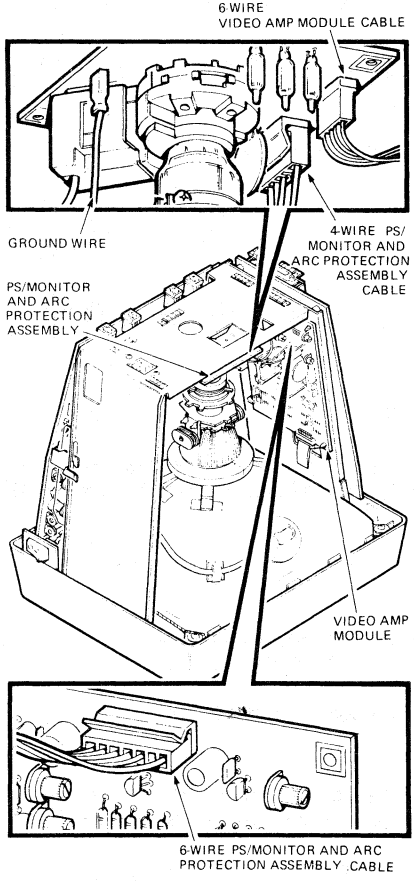
Rear enclosure/base assembly (Paragraph 2.4).

WARNING
You must discharge the CRT (Paragraph 2.6) before removing the video amp module.

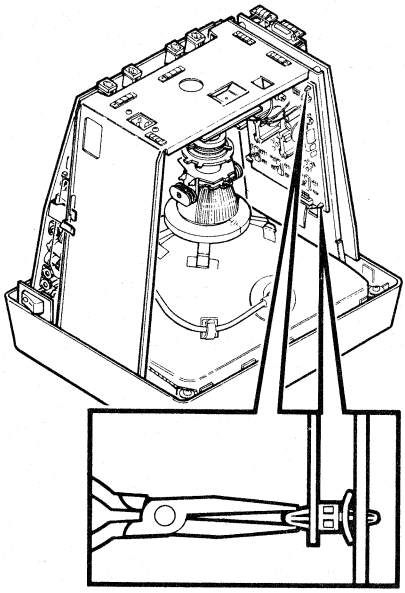
1. Disconnect the following cables on the video amp module.

   - 16-wire cable to the terminal control module
   - 6-wire cable to the PS/monitor and arc protection assembly

2. On the neck of the CRT, disconnect the 3-wire video amp cable on the PS/monitor and arc protection assembly.



3. Remove the video amp module by squeezing the module's standoffs with needlenose pliers.



To install the video amp module, reverse steps 1 through 3. The video amp module has two notches on the side. Place these notches on the legs of the chassis.

2.9 AC INLET/SWITCH ASSEMBLY

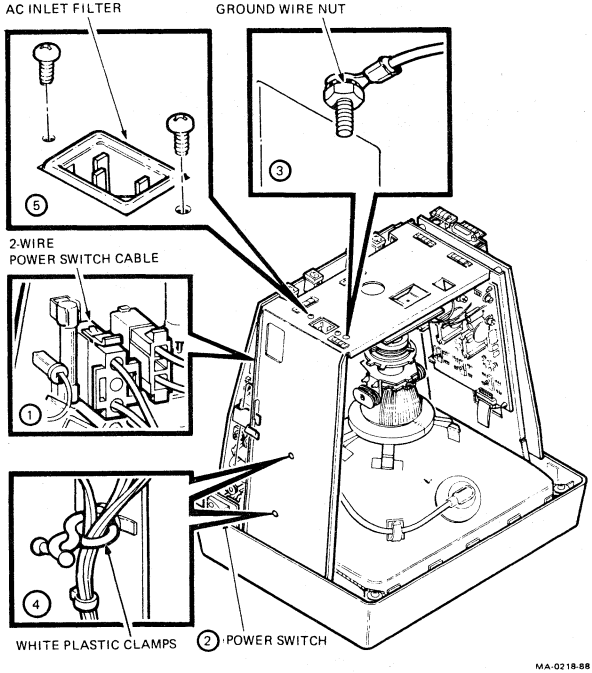
Remove the ac inlet/switch assembly as follows. First remove the following FRUs.

Rear panel (Paragraph 2.2).

Rear enclosure/base assembly (Paragraph 2.4).

1. Find the 2-wire cable from the power switch to the PS/monitor and arc protection assembly. Disconnect the cable at the assembly.
2. Pull the switch up and out of the holder. Bring the switch to the front of the unit.

3. Use a 3/8 inch nutdriver to remove the nut that holds the ground wires from the ac inlet filter to the chassis.
4. Release the ac inlet/switch assembly cable from white plastic clamp on the side of the chassis.
5. Hold the ac inlet filter while removing the two phillips screws on the rear chassis. Remove the filter.



To install the ac inlet/switch assembly, reverse steps 1 through 5.

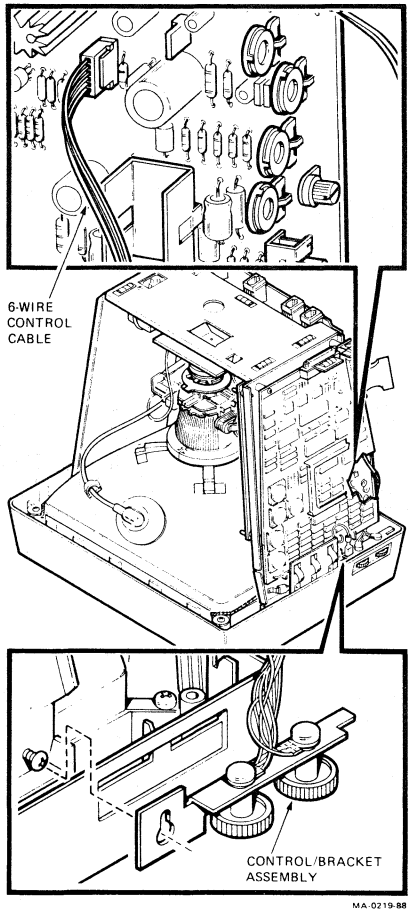
2.10 CONTROL/BRACKET ASSEMBLY (BRIGHTNESS AND CONTRAST)

Remove the control/bracket assembly as follows. First remove the following FRUs.

Rear panel (Paragraph 2.2).

Rear enclosure/base assembly (Paragraph 2.4).

1. Disconnect the 6-wire control cable on the PS/ monitor and arc protection assembly by pressing the tab under the connector.
2. Loosen the phillips screw that holds the brightness and contrast controls to the chassis.
3. Lift the control/bracket assembly off the chassis.



To install the control/bracket assembly, reverse steps 1 through 3.

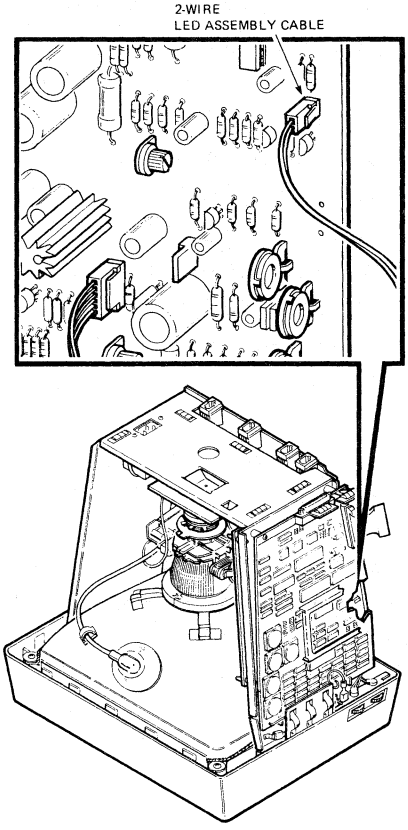
2.11 LED ASSEMBLY (POWER INDICATOR)

Remove the LED assembly as follows. First remove the following FRUs.

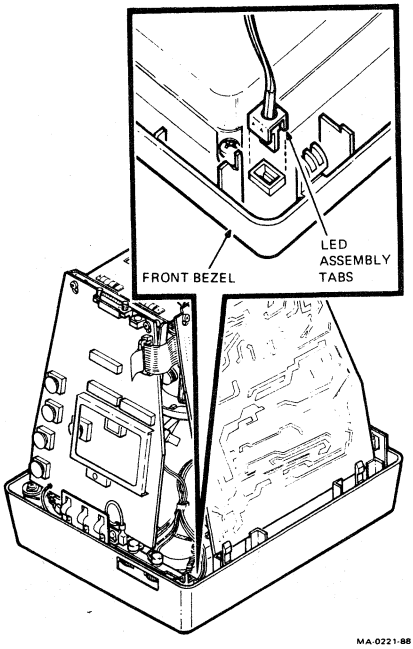
Rear panel (Paragraph 2.2).

Rear enclosure/base assembly (Paragraph 2.4).

1. Find the 2-wire cable from the front bezel to the PS/monitor and arc protection assembly. Disconnect the cable at the module by pressing the tab under the connector.



2. Lift the LED assembly from the front bezel by spreading the tabs on the LED with a small-bladed screwdriver.



To install the LED assembly, reverse steps 1 and 2.

2.12 CRT/BEZEL/CHASSIS ASSEMBLY

The CRT/bezel/chassis is one FRU. Do not try to remove or repair any part of this assembly in the field. To replace the assembly, remove and save the following FRUs.

WARNING
You must discharge the CRT (Paragraph 2.6) before removing the PS/monitor and arc protection assembly, and the video amp module.

Rear panel (Paragraph 2.2)

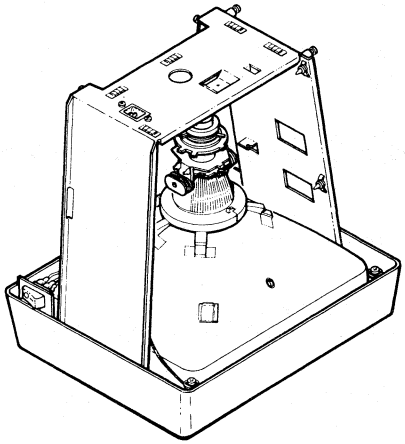
ROM cartridge (Paragraph 2.3).

Rear enclosure/base assembly (Paragraph 2.4)

Terminal control module (Paragraph 2.5)

PS/monitor and arc protection assembly (Paragraph 2.7)

Video amp module (Paragraph 2.8)



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To install the CRT/bezel/chassis assembly, reverse the above procedures. Place a piece of paper on a flat working surface. Carefully place the new monitor facedown on the paper. The paper prevents scratches to the monitor bezel.

2.13 CRT DISPOSAL (DIGITAL FIELD SERVICE ONLY)

This section describes how to safely dispose of the monitor's cathode-ray tube (CRT).

> **NOTE**
> **This procedure supersedes all other tech tips about replacing and disposing of CRTs. This procedure is for Digital personnel only, and is not intended for use by OEM and self-maintenance customers.**

CRTs are glass vacuum tubes. Because air pressure outside the tube is greater than air pressure inside, there is always the possiblity of accidental implosion.

> **WARNING**
> **You must handle CRTs very carefully to avoid accidental implosion and shattering glass.**

To prevent personal injury from CRT implosion, use the following guidelines and disposal procedure to remove and dispose of a CRT. These guidelines and procedure are Digital field policy for all CRTs more than 3 inches in diameter.

Location
Work in areas where risks and exposure are limited to trained Digital personnel. Only Digital personnel should be in the area during CRT removal and replacement.

Protective Gear
Any person replacing a CRT must wear at least safety goggles and the approved gloves.

Safety goggles	pn 29-16141-00
Gloves	pn 29-16146-00

> **WARNING**
> **To avoid injury to the eyes or hands, always wear goggles and gloves when you work with a CRT.**

Handling a CRT

- Never handle the CRT by the neck. Always use two hands and hold the CRT by the sides near its face.
- Keep the CRT away from your body during handling.
- Do not let the neck strike anything.
- Do not rest the CRT on its neck.
- Do not let the CRT touch any tools, such as screwdrivers and soldering irons.

Stocking and Storage

All CRTs must be kept in a closed container or mounted in the device cabinetry.

CRT Disposal

Use the following procedure to safely dispose of CRTs. Always perform this procedure at a Digital facility.

> **WARNING**
> **Do not dispose of any CRT until it is rendered inoperative and safe to dispose.**
>
> **Never perform the following disposal procedure at the customer site. Return the defective CRT to the local Digital facility for disposal.**

At the Digital facility you will need

- An area clear of nonessential personnel
- A second person in the area in case of an emergency
- Safety goggles (PN 29-16141-00)
- Gloves (PN 29-16146-00)
- Pliers (PN 29-10311-00)

> **WARNING**
> **Be careful when performing this procedure. For your safety, you must use the specified gloves and goggles.**

Never handle pieces of phosphor-coated glass without wearing protective gloves.

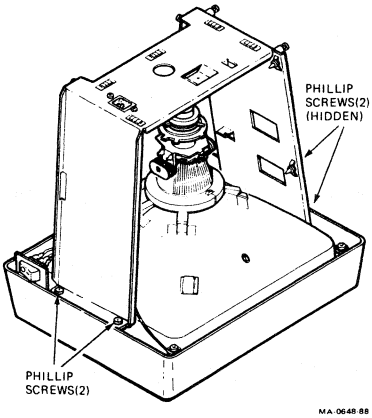
Before you perform the following procedure, remove the FRUs listed in Paragraph 2.12.

Then, remove the following FRUs.

Inlet/switch assembly (Paragraph 2.9)

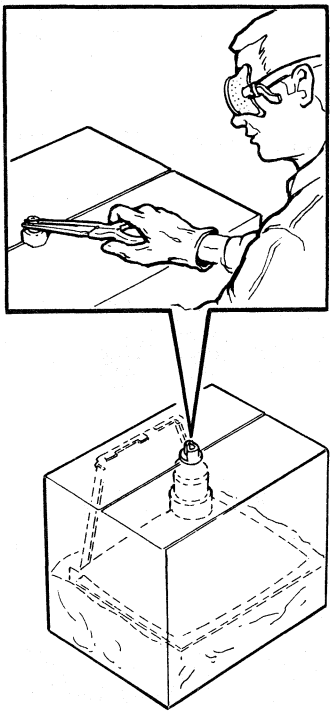
Control/bracket assembly (Paragraph 2.10)

1. Remove the four phillips screws that secure the chassis to the CRT/bezel.



2. Remove the chassis.
3. Discard the chassis.
4. Place the old CRT and the original packing material in the container from which you removed the new CRT.
5. Seal up the container so that only the very tip of the CRT neck is exposed.
6. Using the specified pliers, slowly crush, but do not snap, the evacuation point. Do not move or disturb the CRT until the hissing sound of inrushing air has stopped.

NOTE
The evacuation point is a protrusion that extends from the circular area defined by the CRT neck pins. The glass protrusion is sometimes encased in a protective plastic cap, and more force is required to crush it.



MA-1500-87

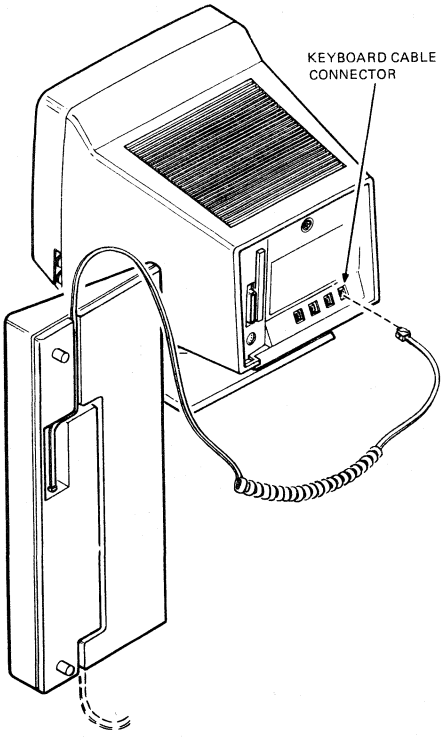
7. Seal the carton with packing tape and dispose of it in the Digital site's trash compactor or receptacle.

NOTE
This safe "gassing" of the CRT is necessary to prevent liability and safety problems that may arise from accidental CRT implosion.

2.14 KEYBOARD

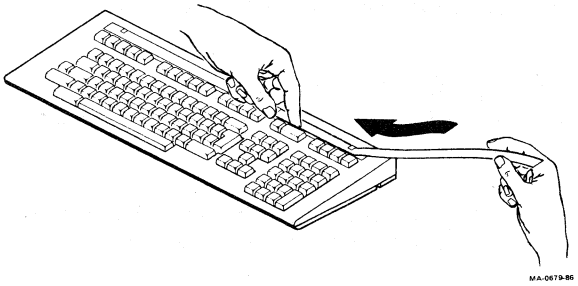
If the keyboard is faulty, replace the complete keyboard (whole option swap). Remove the keyboard as follows.

1. Disconnect the keyboard cable from the rear of the terminal.



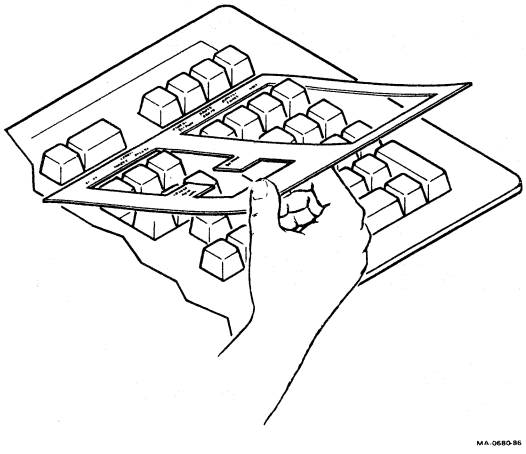
Install the legend strip (non-U.S./U.K. keyboards only).

1. Remove the legend strip from the old keyboard.
2. Slide the legend strip under the tabs, so it fits over the legend strip already on the new keyboard.

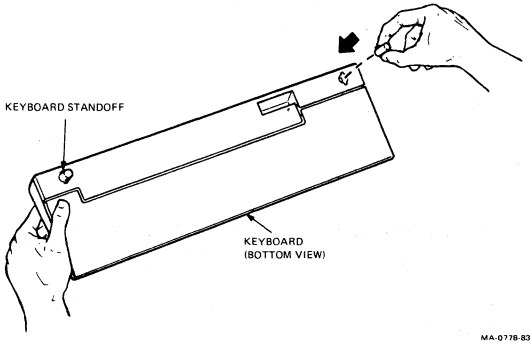


Install the template for the local editing keys (non-U.S./U.K. keyboards only).

1. Place the template so that it fits around the editing and numeric keypads.

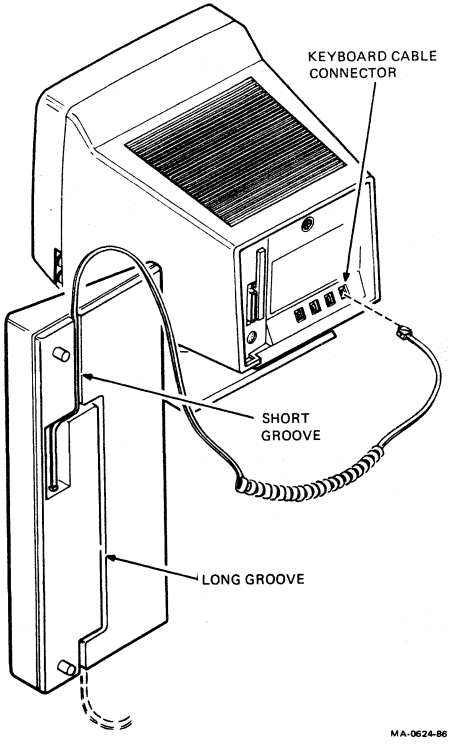


Install the keyboard standoffs.



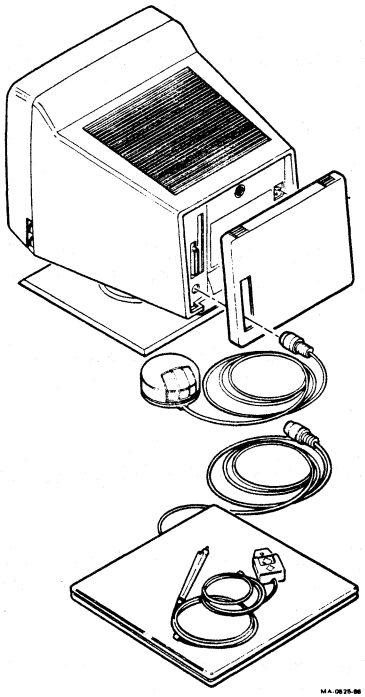
Connect the keyboard to the terminal.

1. Turn the terminal's power switch off (0).
2. Insert the long straight end of the keyboard cable into the connector on the bottom of the keyboard.
3. Press the cable into the long or short groove. Use the long groove if routing to the left. Use the short groove if routing to the right.
4. Insert the other end of the cable into the keyboard connector on the rear of the terminal.



2.15 MOUSE or TABLET (OPTIONS)

The VSXXX-AA mouse and the VSXXX-AB graphics tablet are single FRUs (whole option swap). The cable for the mouse or tablet connects to the 7-pin locator device connector on the rear of the terminal. To disconnect a mouse or tablet, first remove the terminal's rear panel (Paragraph 2.2). Then disconnect the mouse or tablet cable from the terminal.



To install a new mouse or tablet, first feed its cable through the opening in the rear panel. Connect the cable to the 7-pin connector, then install the rear panel.

NOTE
The cable for the mouse or tablet has a keying plug. Make sure you align the keying plug to the slot on the 7-pin connector.

3 ALIGNING THE VIDEO MONITOR

3.1 GENERAL

This chapter shows you how to align the VT340 monitor. You do not have to perform every adjustment procedure each time you align a monitor. However, you should check all adjustments in the order shown, because many adjustments affect each other. If a setting is already correct, you can skip that adjustment and go on to the next one.

You must use the screen alignment test patterns to make all adjustments. Use Paragraph 3.2.1 to set up the terminal for adjustments.

Use a metric measuring tape (PN 29-25342-00) to measure the dimensions of the screen display. To avoid scratching the screen with the tape's metal clip, start the measurement at 10 cm (100 mm). Make sure all adjustments are made under these conditions.

3.2 MONITOR ADJUSTMENTS

Figure 3-1 shows the location of the controls used to make most adjustments. The individual procedures describe the purpose of each control.

Let the terminal warm up for at least 20 minutes before performing any adjustments (color bias and drive controls) on the video amp module.

Let the terminal warm up for at least 5 minutes before performing any other adjustments on the PS/monitor and arc protection assembly.

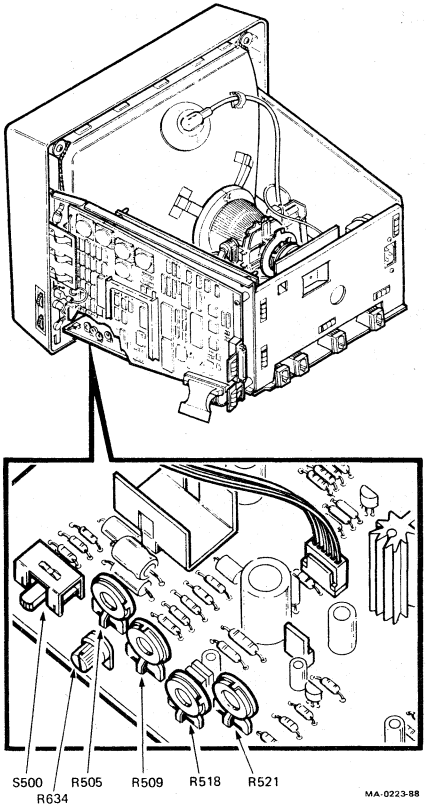


Figure 3-1 Video Adjustment Controls (1 of 2)

S500 Service switch
R634 Subcontrast
R505 Vertical height
R509 Vertical linearity
R518 Vertical centering
R521 Rectangularity

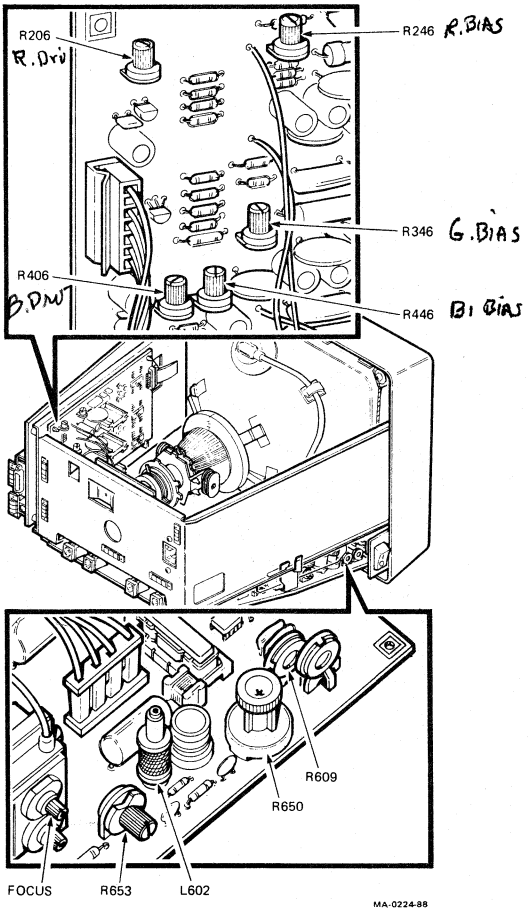


Figure 3-1 Video Adjustment Controls (2 of 2)

R206	Red drive	L602	Horizontal width
R246	Red bias	R609	Horizontal phase
R346	Green bias	R650	Horizontal centering
R406	Blue drive	R653	Cutoff
R446	Blue bias		

3.2.1 Before You Start

Before you perform adjustments, set up the terminal as follows. First remove the following FRUs.

Rear panel (Paragraph 2.2).

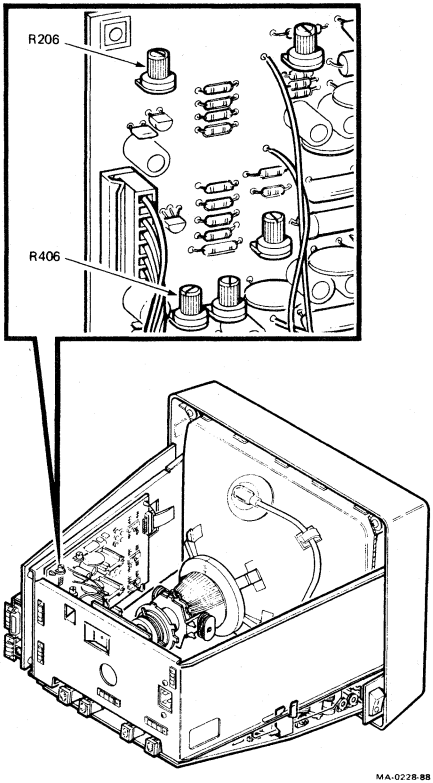
Rear enclosure/base assembly (Paragraph 2.4).

1. Place your terminal on a nonconductive surface.
2. Reconnect the keyboard cable and power cord.
3. Turn the power switch on (1). Wait for "VT340 OK" to appear on the screen.
4. Let the terminal warm up for at least 20 minutes before performing any adjustments (color bias and drive controls) on the video amp module.

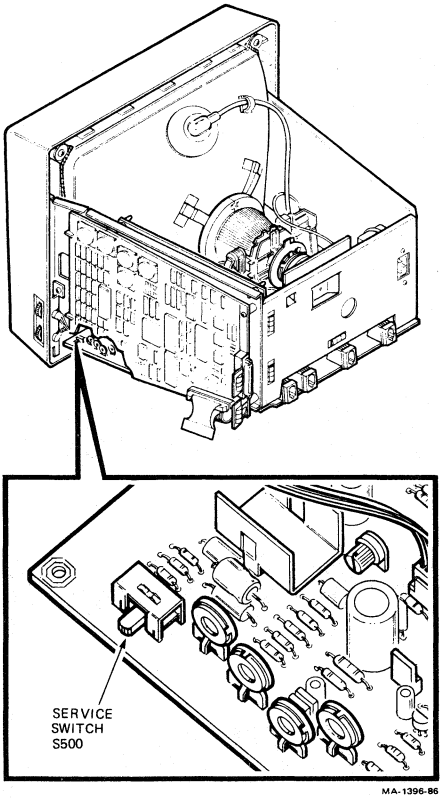
3.2.2 Setting Cutoff

Cutoff is the point of operation in a CRT tube where current does not flow.

1. Set the brightness and contrast controls to minimum.
2. Turn the R246 (R-Bias), R346 (G-Bias), and R446 (B-Bias) controls to their midrange position.
3. Set the R206 (R-Drive) and R406 (B-Drive) controls on the video amp module to a full counterclockwise position.


R206
R406
MA-0228-88

4. Set the service switch (S500) on the PS/monitor and arc protection assembly to the rear position.

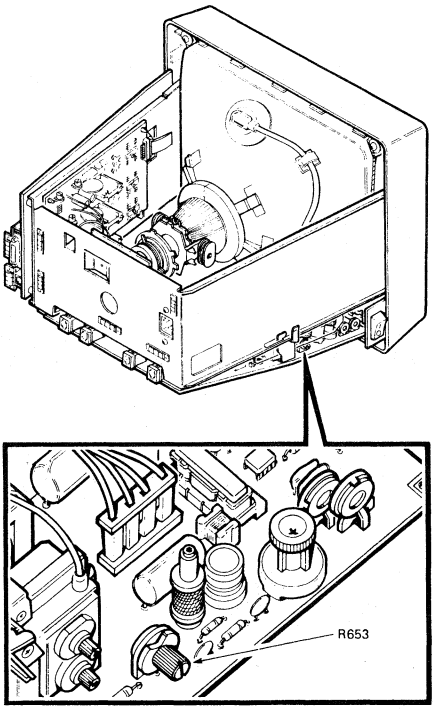


5. Slowly turn the R653 (G2) control clockwise, until you see a horizontal line on the screen. If a line is present, go to step 6. Note the color of the line — red, green, or blue.

CAUTION
Whenever the horizontal line appears, set the line to a dim viewing level. Failure to do so may burn the phosphor on the CRT.

6. Slowly turn the R653 (G2) control counterclockwise, *until the line just disappears.* Do not turn the control too far.

   The cutoff point for that color is now set. Do not adjust the cutoff (bias) control for that color during the rest of this procedure.



NOTE
When you set the bias controls for the other two colors, make sure the horizontal line just disappears. It is important to set all three bias controls so that the three colors disappear at the same point.

7. Turn the bias control for one of the two other colors, until you see a line of that color on the screen. Then turn the control back, until the line just disappears.
8. Repeat step 7 for the third color.
9. Set the service switch (S500) on the PS/monitor and arc protection assembly to the forward position.
10. Turn the contrast control to maximum.
11. Turn the brightness control until the raster (white lines) just appears. Then turn the control back, until the raster just disappears.

NOTE
Any data now displayed on the screen will be green.

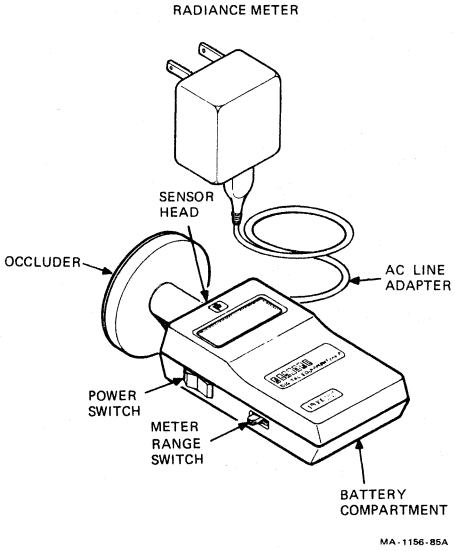
3.2.3 Displaying Screen Tests

The screen test displays a series of test patterns. You run the screen test from the Diagnostic Set-Up screen (Paragraph 1.4).

1. Let the terminal warm up for at least 20 minutes before performing adjustments (color bias and drive controls) on the video amp module.
2. Press **Set-Up** to enter set-up. The Set-Up Directory screen appears.
3. Use the arrow keys to move the highlighting cursor to the **Diagnostics Set-Up** field. Press **Enter** to display the Diagnostics Set-Up screen.
4. Use the arrow keys to select **Run Screen Tests** in the Diagnostics Set-Up screen.
5. Press **Enter** to run the screen tests. The monitor displays a series of screen tests repeatedly. To keep a test pattern on the screen, press the **Hold Session** key.

3.2.4 Color Gain

For this adjustment, you need the radiance meter supplied with the color video service kit (PN A2-S0099-01). The kit comes with a manual that contains important information on using the meter. You should read this manual before using the meter.



Prepare the meter for use by removing the cap and attaching the occluder assembly.

CAUTION
Do not touch the filter while the cap is removed. Do not tighten the occluder too far.

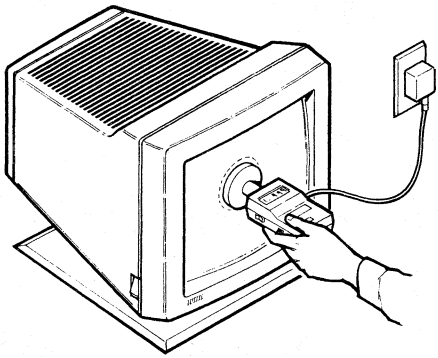
The meter's power switch has three positions.

Down	Power on, without a backlight
Center	Power off
Up	Power on, with a backlight

Set the power switch to the down position (power on without a backlight).

The meter range switch lets you select the accuracy of the meter's decimal readout. Set the switch to the second position from top of meter (1.999).

Place the occluder firmly against the center of the screen. Take a red china pencil and draw an arc on the screen around the occluder.

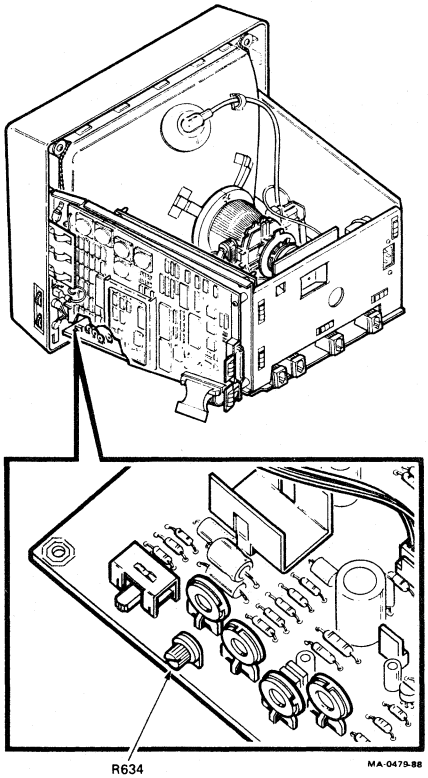


MA-0175-87

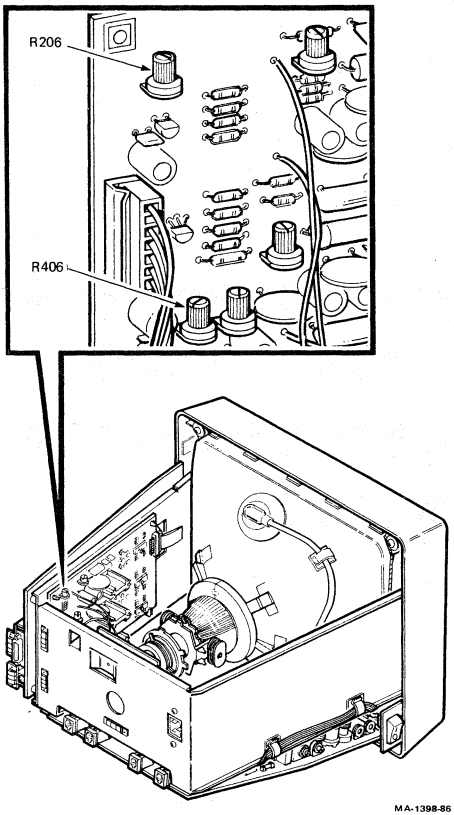
NOTE
You must take all meter readings with the meter centered in this arc.

1. Run the screen test.
2. Stop the test at the green screen by pressing **Hold Session.**
3. Ensure that the contrast control is set to maximum and the brightness control is adjusted so that the raster just disappears.
4. Place the occluder in the arc on the screen and adjust R634 (sub-contrast control) until the meter reads 0.095.

5. Continue the screen test by pressing **Hold Session**. Stop the test at the blue screen by pressing **Hold Session** again.



6. Place the occluder in the arc on the screen and adjust the R406 (B-Drive) control until the meter reads 0.192.

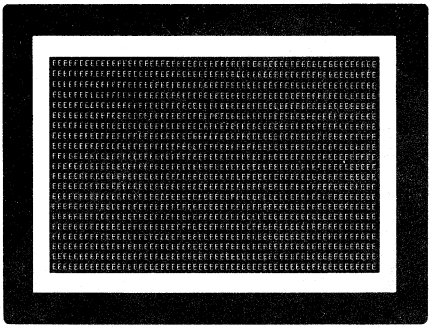


7. Continue the screen test by pressing **Hold Session**. Stop the test at the red screen by pressing **Hold Session** again.
8. Place the occluder in the arc on the screen and adjust the R206 (R-Drive) control until the meter reads 0.153.
9. Display the white screen, using **Hold Session**.
10. Adjust the brightness and contrast controls to a comfortable viewing level.

3.2.5 Display Size

This section describes how to check and adjust horizontal centering, horizontal phase, rectangularity, vertical centering, horizontal width, and vertical height. To perform these adjustments, first do the following.

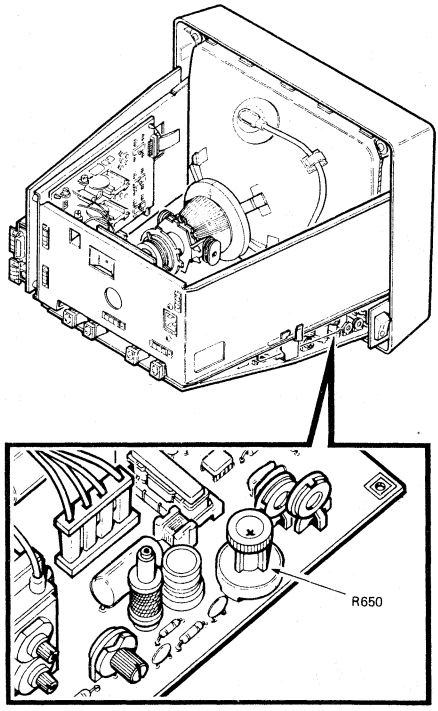
1. Run the screen test (Paragraph 3.2.3).
2. Stop the test at the screen of E's with a white border, by pressing **Hold Session**.



MA-0629-86

3.2.5.1 Horizontal Centering - Check and adjust the horizontal centering as follows.

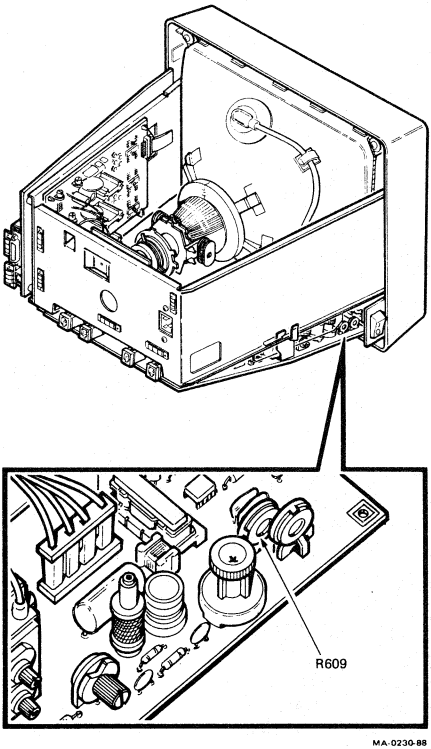
1. Set the brightness control to maximum. Then set the contrast control to minimum.
2. Check the raster. It should appear centered on the screen.
3. If necessary, adjust the R650 (Hor. Center) control to center the raster.



MA-0229-88

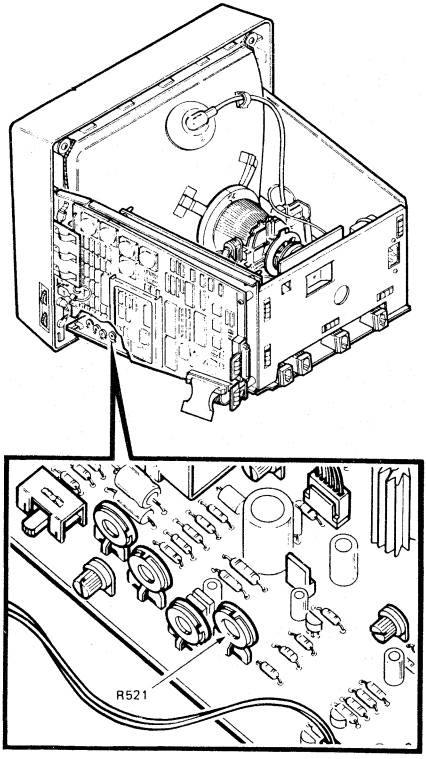
3.2.5.2 Horizontal Phase - Check and adjust the horizontal phase as follows.

1. Adjust the brightness control until the raster appears around the screen of E's with a white border.
2. Adjust the R609 (H-Phase) control until the display appears centered in the raster.



3.2.5.3 Rectangularity - This procedure ensures that the sides of the video display are straight, not curved. The curved line effect is known as *pin-cushioning* (when the lines curve in) or *barreling* (when the lines curve out).

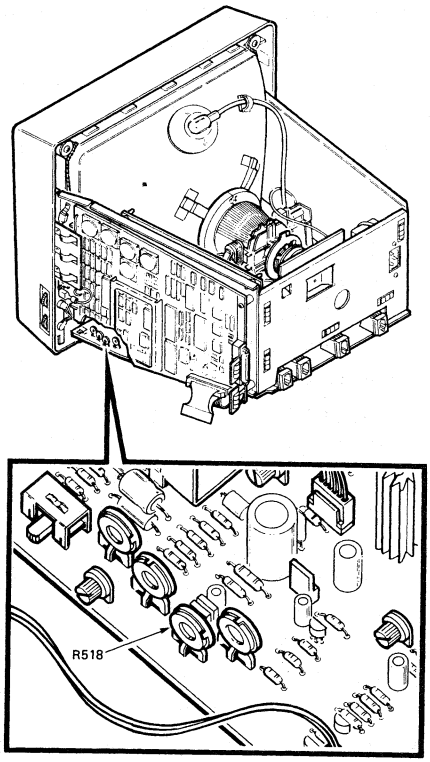
1. Check that the sides of the screen of E's are straight.
2. If necessary, adjust the R521 (Side-Pin) control until the sides appear straight.



MA-0231-88

3.2.5.4 Vertical Centering - Check and adjust the vertical centering as follows.

1. Look at the screen of E's. It should appear vertically centered on the screen.
2. If necessary, adjust the R518 (V-Cent) control until the display appears centered.

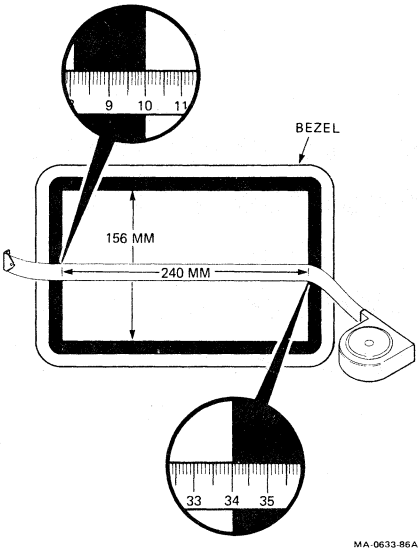


MA-0232-88

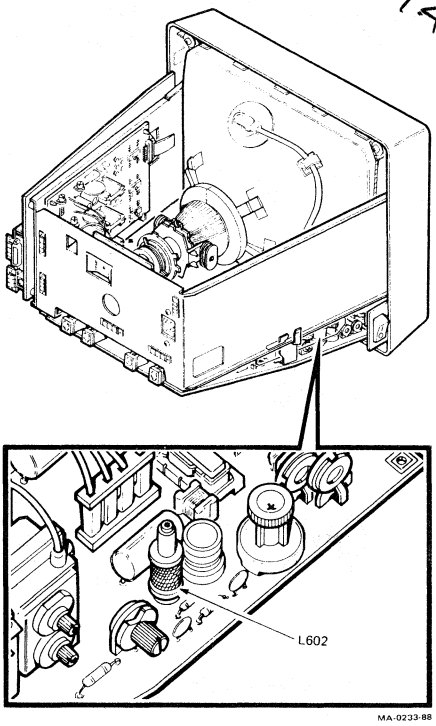
3.2.5.5 Horizontal Width and Vertical Height - Check and adjust the horizontal width and vertical height as follows.

NOTE
In the following steps, use a metric measuring tape (PN 29-25342) to measure the dimensions of the screen display. To avoid scratching the screen with the tape's metal clip, start each measurement at 10 cm (100 mm).

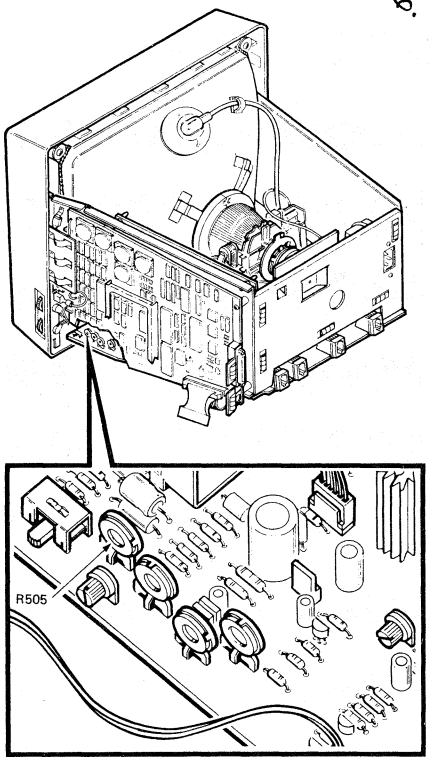
In the following figure, the screen of E's display is removed for clarity.



1. Check the outside edges of the screen of E's for a horizontal width of about 240 mm ± 2.0 mm.
2. If necessary, adjust the width to 240 mm by using the L602 (horizontal width) control.



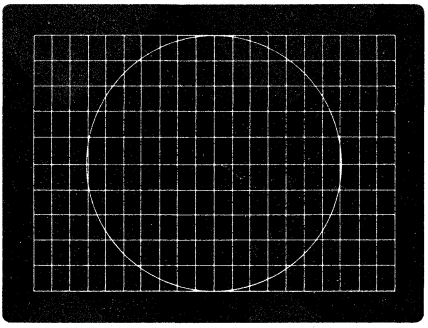
3. Check the outside edges of the screen of E's for a vertical height of 156 mm ± 2.0 mm.
4. If necessary, adjust the height to 156 mm by using the R505 (V-Size) control.



3.2.6 Character Quality

This section describes how to check and adjust linearity and focus. To perform these adjustments, first do the following.

1. Run the screen test (Paragraph 3.2.3).
2. Stop the test at the following screen by pressing the **Hold Session**.



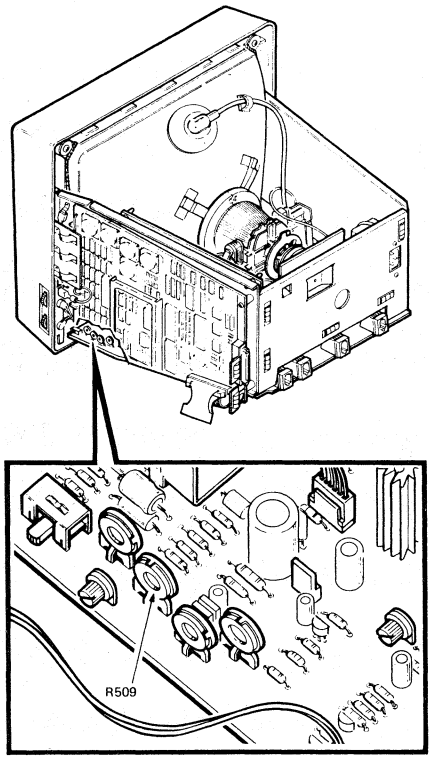
MA-0636-86

3.2.6.1 Linearity - Check and adjust the linearity as follows.

1. Check the height of the four corner blocks in the test display. The blocks should all be the same height.

   Also check if the circle appears round.

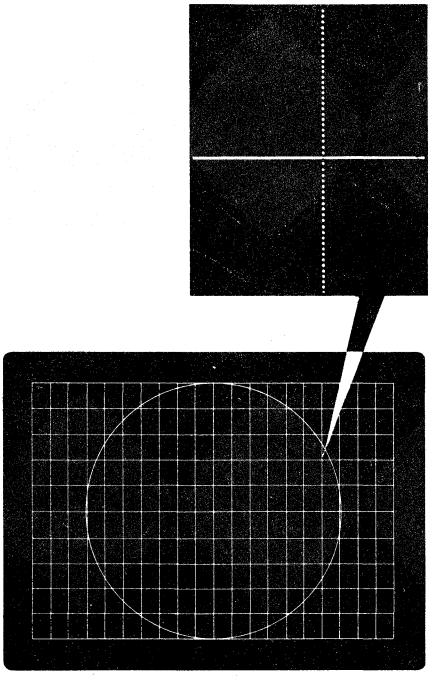
2. If necessary, adjust the R509 (V-Lin) control so the blocks are the same height and the circle appears round.



3. If you adjust the linearity, you must readjust the vertical height (Paragraph 3.2.5.5).

3.2.6.2 Focus - Check and adjust the focus as follows.

1. Find the point on the crosshatch pattern that is four blocks from the right edge and three blocks from the top.

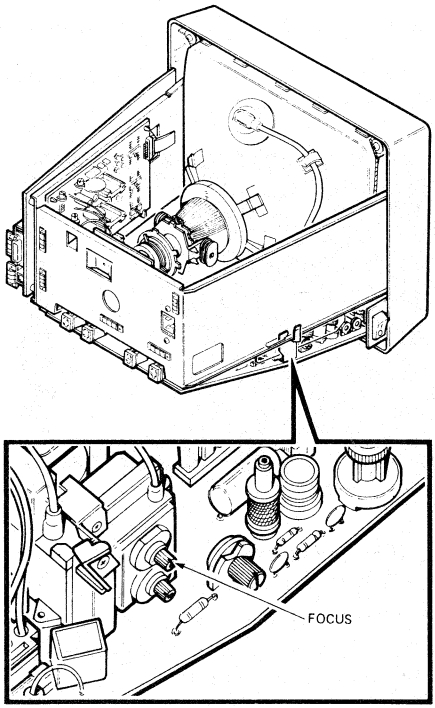


MA-0636-86A

2. Examine the intersecting lines for sharpness.

NOTE
Make sure the screen is clean. A dirty screen can appear to affect the focus. In some cases, the operator may want the focus set to suit personal preference.

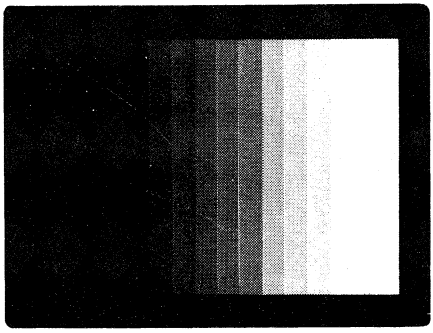
3. If necessary, adjust the focus control on the fly-back transformer so the intersecting lines are sharp.



3.2.7 Viewer Preference

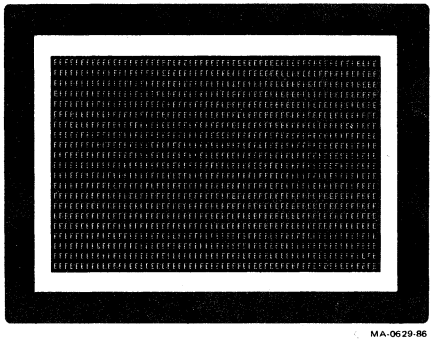
This procedure lets you set the brightness and contrast to the viewer's preference.

1. Run the screen test (Paragraph 3.2.3).
2. Stop the test at the screen with a gray scale by pressing **Hold Session**.



MA 0052 87

3. Adjust the brightness and contrast controls to the preferred viewing level.
4. Continue the screen test by pressing **Hold Session**. Stop the test at the screen of E's by pressing **Hold Session** again.



MA-0629-86

5. Check the horizontal width and vertical height (Paragraph 3.2.5.5). Adjust if necessary.

4 USING SET-UP

4.1 GENERAL

The VT340 terminal has a set-up utility that lets you change the settings of operating features from the keyboard. This chapter provides a brief summary of how to use set-up. Chapter 5 of the *Installing and Using the VT330/VT340 Video Terminal* describes the set-up utility in detail.

4.2 WHAT IS SET-UP?

The set-up utility is a series of display screens. Each screen lists a group of related features, such as communications or diagnostics. Each set-up screen has basic directions at the bottom for selecting and changing feature settings. You can move from screen to screen in set-up.

4.2.1 Dual Sessions

The VT340 can run two sessions with the host system at the same time. Each session has its own set-up settings. If the terminal is set for dual sessions, the set-up screens display the settings for the active session. Chapters 3 and 8 of *Installing and Using the VT330/VT340 Video Terminal* describe how to set up and use dual sessions.

4.3 ENTERING AND LEAVING SET-UP

You enter set-up by pressing the Set-Up key on the top row of the keyboard. When you press Set-Up, the Set-Up Directory screen appears (Figure 4-1). The top half of the Set-Up Directory lists all the VT340 set-up screens. The bottom half lists some action you can perform directly from the Set-Up Directory.

You can leave set-up from any set-up screen, by pressing Set-Up again.

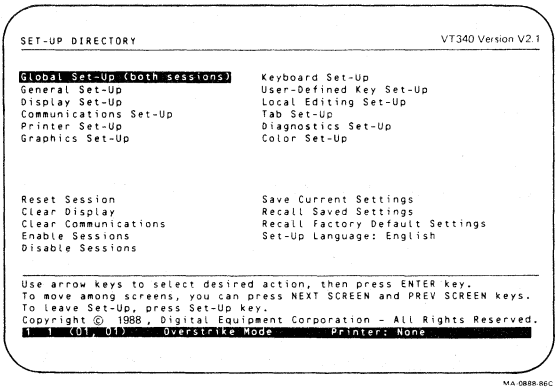


Figure 4-1 Set-Up Directory

4.4 SELECTING A SET-UP SCREEN

You can select any set-up screen from the Set-Up Directory screen.

1. Press Set-Up to enter set-up. The Set-Up Directory appears.
2. Use the arrow keys to move the highlighting cursor to the name of the screen you want to display.
3. Press Enter. The selected screen appears.

You can also use the **Next Screen** key to move from one set-up screen to the next. You can use the **Prev Screen** key to back up to a screen you have already seen. The screens appear in the order listed in the Set-Up Directory. You can press **Next Screen** and **Prev Screen** when viewing any set-up screen.

To return to the Set-Up Directory, press the **Select** key.

4.5 SELECTING SET-UP FEATURES

You use the arrow keys to move the cursor to any field on a set-up screen. There are two basic types of fields.

Action fields	When you move to an action field and press **Enter**, the terminal performs that action. The Set-Up Directory, User-Defined Key, and Tab screens have action fields.
Features	Features have more than one setting. You use the →, ←, or **Enter** key to change the current setting of a feature. When you leave set-up, the terminal uses the new setting you selected.

Do not change any settings unless necessary.

4.6 SAVING SET-UP VALUES

You may want to save the current settings in set-up before you run tests. Check with the user. When you save settings, you can recall them later. You can save all set-up settings as follows. If the terminal is set for dual sessions, this procedure only applies to the active session.

1. Press **Set-Up** to enter set-up. The Set-Up Directory screen appears.
2. Use the arrow keys to move to the **Save Current Settings** field in the Set-Up Directory screen.
3. Press **Enter** to save the current settings.

4.7 RECALLING SET-UP VALUES

You can reset the terminal to a previous set of set-up values. There are two types of set-up values you can recall. Check with the user to see which settings were in use.

- Factory-default settings
- Saved settings (previously selected by the user)

You can recall set-up values as follows. If the terminal is set for dual sessions, this procedure applies only to the active session.

1. Press **Set-Up** to enter set-up. The Set-Up Directory screen appears.
2. Use the arrow keys to select the **Recall Factory Default Settings** or **Recall Saved Settings** field.
3. Press **Enter** to recall the selected set-up values.

A VT330/VT340 DOCUMENTATION

In addition to this guide, you can order the following VT330/VT340 documents from Digital.

Installing and Using
The VT330/VT340 Video Terminal **EK-VT3XX-UG**
This manual provides users with the information needed to install, operate, and maintain the VT330/VT340 terminals. The manual describes all controls, indicators, and keys. The manual also describes set-up and special features such as dual sessions and local editing.

VT330/VT340
Programmer Reference Manual (two volumes)

Vol. 1: Text Programming **EK-VT3XX-TP**
This manual provides programmers with information on character processing, character codes, and control functions available for general text applications.

Vol. 2: Graphics Programming **EK-VT3XX-GP**
This manual provides programmers with information on graphics features of the VT330 and VT340 terminals. The manual describes three types of graphics programming: ReGIS, Tektronix 4010/4014, and sixels. The manual also describes how to program a locator device (mouse or graphics tablet) and how to print graphics.

VT330/VT340
Programmer Pocket Guide **EK-VT3XX-HR**
This guide is a summary of the text and graphics control functions described in the two-volume programmer reference manual.

VT330 Pocket Service Guide **EK-VT330-PS**
This guide provides service personnel with the information needed to test, troubleshoot, and repair the VT330 monochrome video terminal.

VT330 Video Terminal IPB **EK-VT330-IP**
VT340 Video Terminal IPB **EK-VT340-IP**
These documents provide a detailed parts breakdown of the VT330 and VT340 field replaceable parts. The documents do not contain part numbers for components on the printed circuit boards. However, these components are listed in the *Field Maintenance Print Set*, ordered separately.

VT330 Field Maintenance Print Set **MP-02411-01**
VT340 Field Maintenance Print Set **MP-02412-01**
These documents provide a complete set of electrical and mechanical schematic diagrams for the VT330 and VT340 terminals.

Installing and Using
The Session Support Utility **AA-JB84B-TE**
This guide provides system managers with instructions for installing SSU software to support dual sessions over a single communication line. The guide is part of an SSU software kit. The kit is available on different media.

_Magtape	00-Q*ZAV-HM
RX50	00-Q*ZAV-H3
TK50	00-Q*ZAV-H5

* = processor number.

B TROUBLESHOOTING THE SESSION SUPPORT UTILITY

The VT340 can run dual sessions over a single communication line if Digital's SSU software is installed on the host system. If you have problems running dual sessions with SSU software, first check the software versions running on the system and the terminal.

- The host system must have VMS V4.4 or later, and SSU V1.0 or later.
- The host system must have VMS V4.4 or later, and SSU V1.0 or later. See "Restrictions" at the end of the appendix.

Next, make sure the following set-up features are set correctly.

In Global Set-Up

- **Dual Terminal** should be "enabled".
- **Terminal Comm Ports** should be set to "Sessions Comm1" or "Sessions Comm2", depending on the comm port you are using.

In Display Set-Up

- **Vertical Coupling** should be "enabled".

The rest of this appendix lists symptoms and solutions for SSU problems. Many of the symptoms shown are DCL error messages. In order to make your error messages appear in the same format as those shown here, type the following command at the DCL $ prompt.

```
$ set message/nofacility/severity/noid/text
```

PROBLEMS WITH ENABLING SSU SOFTWARE

When SSU software is installed correctly, you can enable it by typing "ssu enable" after the $ prompt. This section lists some error messages that may appear on the terminal when you try to enable SSU software.

Message

```
$ ssu enable
%E, failure to enable multiple sessions on <dev:>
%E, specified device is not associated with a
    virtual terminal.
```

Solution

- This error is often caused by using the DCL command SET HOST to log into the host system. When you use SET HOST, the system identifies the terminal as an RTxx: device. SSU does not work on RTxx: devices. You must log directly onto the system.
- Use the DCL command SHOW TERMINAL to see if your system has virtual terminals enabled. The first line of the response should show "Terminal: _VTAnnn". If not, this is an SSU problem that needs to be resolved on the host system. Have the system manager do an SSU INSERT on the physical line. See *Installing and Using the Session Support Utility.*

Message

```
$ ssu enable
%E, error opening device TDA:
%W, no such device available
```

Solution

- Make sure (1) SSU was installed without any errors, and (2) the system startup procedure executes the file called SSU_STARTUP.COM. See *Installing and Using the Session Support Utility.*
- Make sure TDDRIVER is loaded on your VMS system, by typing the DCL command:

  ```
  $ SHOW DEVICE _TDA
  ```

You should get a list of terminal devices in the left column, including at least TDA0: . If the message "No such device available" appears, TDDRIVER is not loaded correctly. This is an SSU problem that needs to be resolved on the host system. See *Installing and Using the Session Support Utility.*

- Make sure the system's TDDRIVER and SSU software have the same version number, and that the version is V1.0 or later. You can check the version numbers with the DCL commands ANALYZE/IMAGE SYS$SYSTEM:TDDRIVER.EXE and ANALYZE/IMAGE SYS$SYSTEM:SSU.EXE. Look for the line saying "Image file identification: SSU V1.0".

Message

```
$ ssu enable
%W, unrecognized command verb - check
    validity and spelling
```

Solution

- Make sure all SSU installation instructions were performed correctly, in particular the ones that include the definition of the SSU command in the common user start-up file. See *Installing and Using the Session Support Utility.*

Message

```
$ ssu enable
!@AJ!@AJ!@AJ!@AJ!@AJ!@AJ
```

Solution
Make sure the set-up features listed at the beginning of this appendix are set correctly.

In Global Set-Up

- **Dual Terminal** should be "enabled".
- **Terminal Comm Ports** should be set to "Sessions Comm1" or "Sessions Comm2", depending on the comm port you are using.

In Display Set-Up

- **Vertical Coupling** should be "enabled".

DECserver LAT PROBLEMS

Symptom
Your session(s) were running fine, but now the terminal is not responding. You think that you are still logged into the system.

Solution

- Select **Disable Sessions** in the Set-Up Directory.
- At the Local> prompt, reconnect to your VMS system and log in. If system parameters allow disconnected jobs, you may now be able to reconnect to your old job.

Symptom
When you try to enable sessions or open a session, the terminal's warning bell sounds or you get thrown back to the Local> prompt.

Solution
Make sure you do not set your terminal server "forward" or "local" characters to Ctrl-\ or Ctrl-T. To be safe, enter the following commands for your terminal server before logging into VMS.

```
Local> set local none
Local> set forward none
```

Restrictions

- DECserver 200 lines
  - For systems running VAX/VMS V4.4 or V4.5: If the LAT+ driver shipped with the DECserver 200 lines is installed, you cannot use SSU on the DECserver 200 lines.

    The normal LAT driver shipped with VAX/VMS V4.4 or V4.5 will work with SSU.
  - For systems running VAX/VMS V4.6 or later and its associated LAT driver shipped with the DECserver 200: You can use SSU on the DECserver 200 lines.
- SSU is not supported on the OPA0: device, due to internal differences between this device and all other terminal devices on VAX/VMS.

C FRU EXPLODED VIEW DRAWINGS

Figures C-1 and C-2 show the VT340 FRUs and their part numbers.

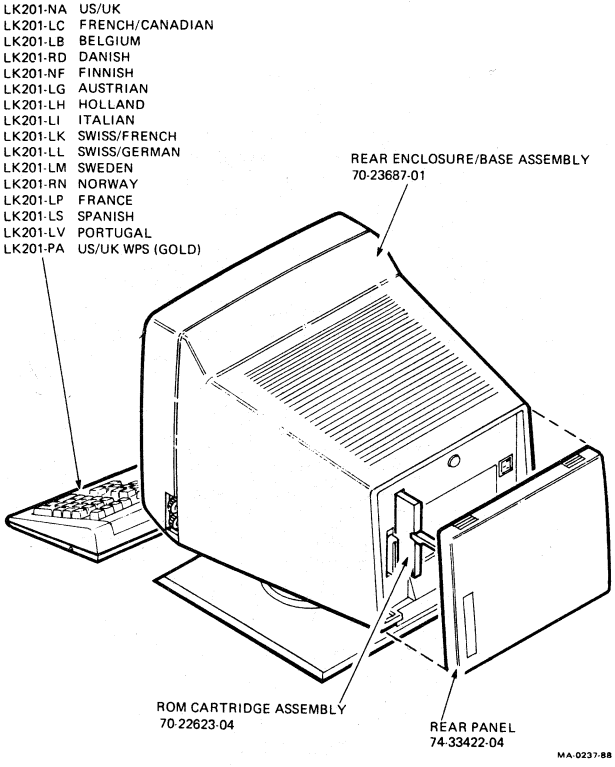


Figure C-1 VT340 FRUs (External View)

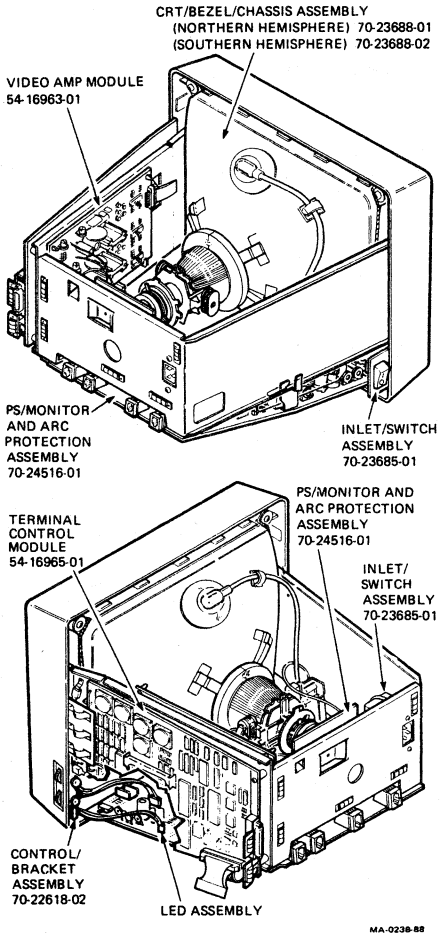


Figure C-2 VT340 FRUs (Internal View)

D CABLES

Figure D-1 shows the possible cable connections for the VT340 terminal. Table D-1 lists the interface cables and their part numbers.

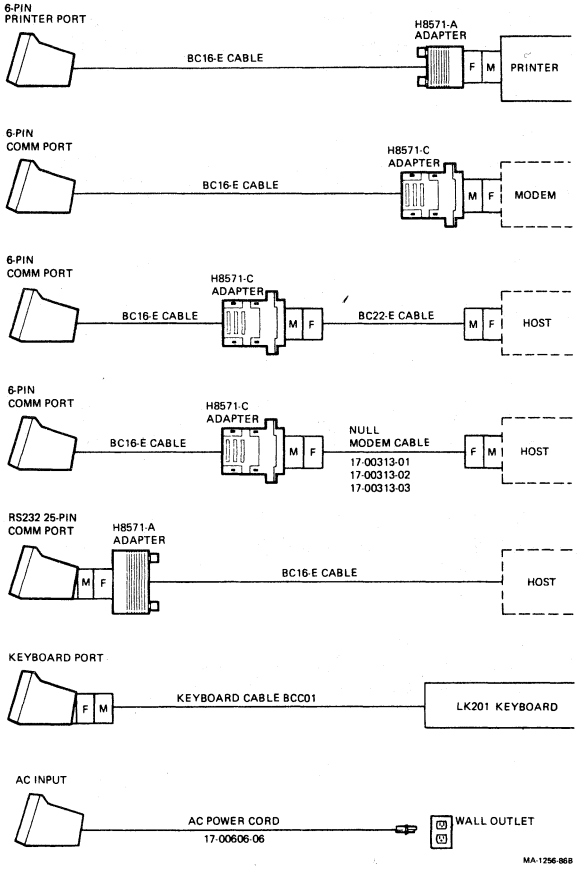


Figure D-1 VT340 Cable Connections

Table D-1 Interface Cables

Part Number	Length	Connector
Adapters		
H8571-A		25-pin F
H8571-C		25-pin M
Extension Cables		
BC22E-10	10 ft (3 m)	25-pin F RS232 to
BC22E-25	25 ft (7.6 m)	25-pin M RS232
Null Modem Cables		
17-00313-01	10 ft (3 m)	25-pin F RS232 to
17-00313-02	25 ft (7.6 m)	25-pin F RS232
17-00313-03	50 ft (15.2 m)	
Communication Cables		
BC16E-10	10 ft (3 m)	6-pin M DEC-423 to
BC16E-25	25 ft (7.6 m)	6-pin M DEC-423
Keyboard Cable		
17-00294-00	6 ft (1.8 m)	Telephone jack
AC Power Cables	**Country**	
17-00198-08	Australia, New Zealand	
17-00199-13	Austria, Belgium, Finland, France, Germany, Netherlands, Norway, Sweden	
17-00606-06	Canada, Japan, Mexico, United States	
17-00310-06	Denmark	
17-00456-09	India, South Africa	
17-00209-09	Ireland, United Kingdom	
17-00457-09	Israel	
17-00364-09	Italy	
17-00210-06	Switzerland	

M = male, F = female.

E PHYSICAL/FUNCTIONAL DIAGRAM

Figure E-1 is a physical/functional diagram showing how the VT340 components work together.

NOTE
All voltage readings are based on a line voltage of 115 Vac. Your readings may differ, depending on your line voltage.

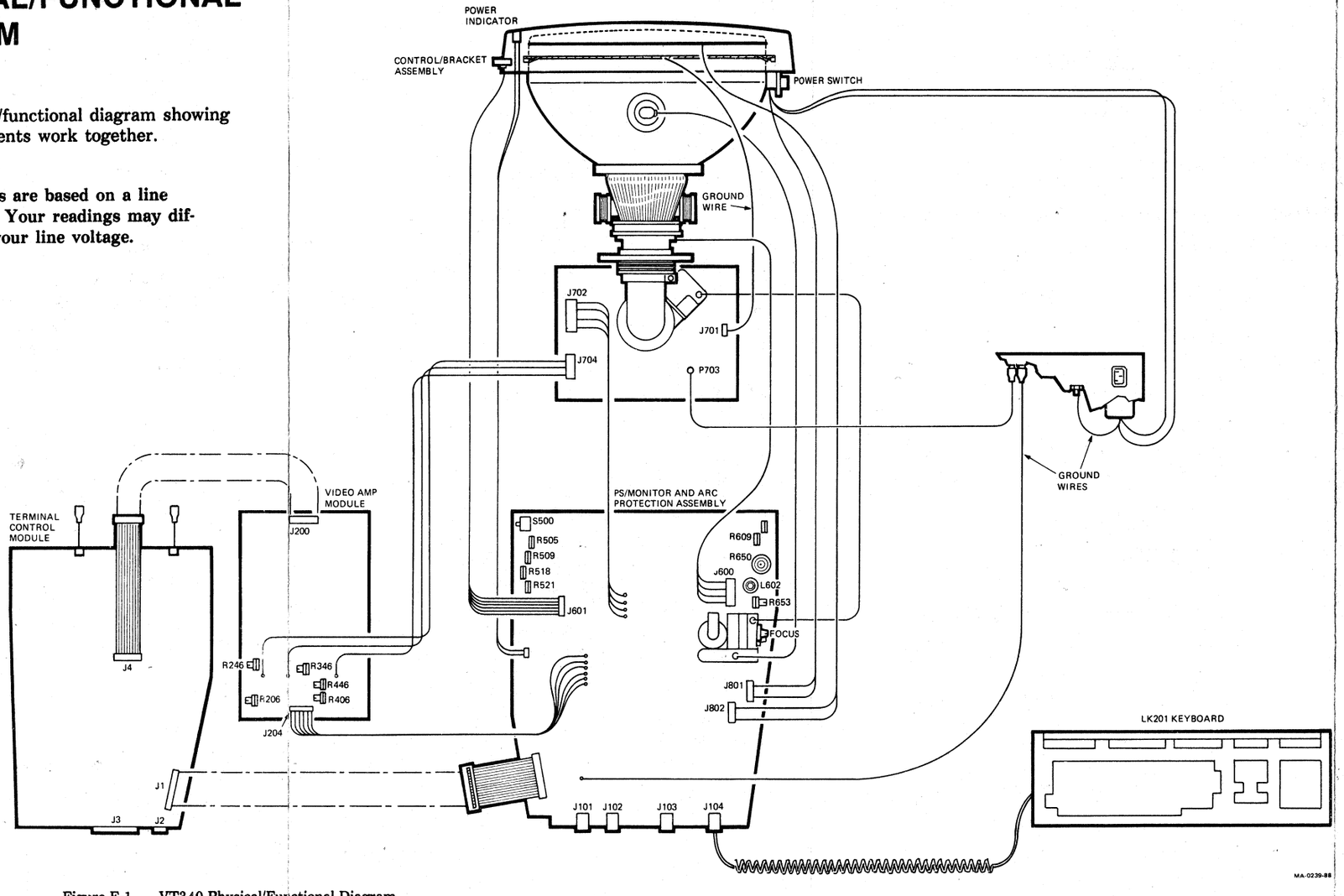


Figure E-1 VT340 Physical/Functional Diagram

PS/Monitor and Arc Protection Assembly

J101 Comm1 Port Connector

Pin			
	1	423DTR	Data terminal ready
	2	423TXD	Transmitted data
	3	GND	Transmit ground
	4	423CMM	Receive ground
	5	423RXD	Received data
	6	423DSR	Data set ready

J102 Comm2 Port Connector

Pin			
	1	H2DTR	Data terminal ready
	2	H2TXD	Transmitted data
	3	GND	Transmit ground
	4	H2CMM	Receive ground
	5	H2RXD	Received data
	6	H2DSR	Data set ready

J103 Printer Port Connector

Pin			
	1	PTRDTR	Data terminal ready
	2	PTRTXD	Transmitted data
	3	GND	Transmit ground
	4	PTRCMM	Receive ground
	5	PTRRXD	Received data
	6	PTRDSR	Data set ready

J104 Keyboard Port Connector

Pin			
	1	K/BTXD	Transmitted data
	2	GND	Ground
	3	+12 V	
	4	K/BRXD	Received data

J600 CRT Yoke

Pin		
	1	Hor yoke
	2	Hor yoke
	3	Vert yoke
	4	Vert yoke

J601 Control/Bracket

Pin		
	1	Contrast
	2	Contrast
	3	Contrast
	4	Brightness
	5	Brightness
	6	Brightness

PS/Monitor and Arc Protection Assembly (Cont)

J701 Ground

Pin			
	1	GND	Ground

J702 Internal PS/Monitor APA

Pin			
	1	GND	Ground
	2	G1	Grid 1
	3	Heater	
	4	Not used	
	5	G2	Grid 2

J704 Video Connector

Pin			
	1	Red	
	2	GND	Ground
	3	GND	Ground
	4	Green	
	5	GND	Ground
	6	Blue	

J801 AC Input

Pin			
	1	Phase	
	2	GND	Ground
	3	Neutral	

J802 Degauss

Pin			
	1	Phase	
	2	Not used	
	3	Neutral	

J900 LED

Pin			
	1	Sig	Signal
	2	-12 V	

Video Amp Module

J200 Terminal Control Connector

Pin			
	5	GNDB	Blue ground
	6	GNDG	Green ground
	11	GNDC	Common ground
	12	GNDR	Red ground

J204 CRT Arc Protection Connector

Pin			
	1	+120 V	
	2	Not used	
	3	+70 V	
	4	+5 V	
	5	ACL	Auto-contrast level
	6	-12 V	
	7	GND	Ground

Terminal Control Module

J1 Power Supply/Monitor Module Interface Connector

Pin			
	1	VERT SYNC	Vertical sync
	2	-12 V	
	3	HOR SYNC	Horizontal sync
	4	+12 V	
	5	RESET	
	6	+5 V	
	7	K/BRXD	Keyboard received data
	8	+5 V	
	9	K/BTXD	Keyboard transmitted data
	10	+5 V	
	11	PTRDSR	Printer data set ready
	12	+5 V	
	13	PTRRXD	Printer received data
	14	+5 V	
	15	PTRCMM	Printer receive ground
	16	+5 V	
	17	PTRTXD	Printer transmitted data
	18	+5 V	
	19	PTRDTR	Printer data terminal ready
	20	GND	Ground
	21	H2DSR	Comm2 data set ready
	22	GND	Ground
	23	H2RXD	Comm2 received data
	24	GND	Ground
	25	H2CMM	Comm2 receive ground
	26	GND	Ground
	27	H2TXD	Comm2 transmitted data
	28	GND	Ground
	29	H2DTR	Comm2 data terminal ready
	30	GND	Ground
	31	E423DSR	Comm1 data set ready
	32	GND	Ground
	33	E423RXD	Comm1 received data
	34	MFGTEST/STATUS	Manufacturing test
	35	423CMM	Comm1 receive ground
	36	+5 V	
	37	423TXD	Comm1 transmitted data
	38	GND	Ground
	39	423DTR	Comm1 data terminal ready
	40	GND	Ground

J2 Locator Connector

Pin			
	1	GND	Ground
	2	MOUSERXD	Received data
	3	MOUSETXD	Transmitted data
	4	-12 V	
	5	+5 V	
	6	+12 V	
	7	Not used	

J3 EIA 25-Pin Comm1 Connector

Pin			
	1	GND	Ground
	2	TXD	Transmitted data
	3	RXD	Received data
	4	RTS	Ready to send
	5	CTS	Clear to send
	6	DSR	Data set ready
	7	SIG GND	Signal ground
	8	CDL	Carrier detect
	9	Not used	
	10	Not used	
	11	Not used	
	12	SPDI	Speed indicator
	13	Not used	
	14	Not used	
	15	Not used	
	16	Not used	
	17	Not used	
	18	Not used	
	19	Not used	
	20	DTR	Data terminal ready
	21	Not used	
	22	Not used	
	23	SPDS	Speed select
	24	Not used	
	25	Not used	

J4 Video Output

Pin			
	5	GND	Ground
	6	GND	Ground
	11	GND	Ground
	12	GND	Ground

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